

Deepwater Riser VIV-Fatigue Case Study

Simulated with CHRONOS — a novel universal bounded-chaos dynamical-systems & fatigue solver

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Executive Summary

This case study analyses vortex-induced-vibration (VIV) driven stress oscillations and fatigue of a deepwater steel catenary riser (SCR) using CHRONOS, a novel universal solver that extends and corrects conventional 3-D chaotic-attractor riser models. Unlike those models — whose 'attractor' actually diverged to $\sim 1e6$ - $1e8$ within ~ 8 s and produced non-physical outputs (e.g. stresses of $4.8e8$ MPa and a fatigue life of $7.1e-20$ cycles) — the CHRONOS Bounded Chaotic Wake-Structure Attractor (BCWSA) is bounded by construction (dissipative trapping region), dimensionally consistent, and built on established VIV physics — a Facchinetti-de Langre-Biolley van der Pol wake oscillator coupled to a Duffing structural oscillator — augmented with a Lorenz-type near-wake convective core that guarantees sustained (non-transient) bounded chaos.

Table 1. Headline results.

Key result	Value
Largest Lyapunov exponent (chaos)	0.915
Kaplan-Yorke fractal dimension	5.081
Correlation dimension D2	1.77
Peak cross-flow A/D	0.98
Dominant bending frequency (Hz)	0.806
Fatigue life — continuous lock-in (years)	0.0046
Fatigue life — design, 15% VIV occurrence (years)	0.0307
Equivalent stress range (MPa)	113.8
Critical current velocity (m/s)	0.5466666666666666
Optimised design life (years)	971
Peak-stress reduction after optimisation (%)	92
Mean structural dissipation (kW)	19.4
Solution bounded (max state)	47.8 (finite)

Case Study Definition and Objectives

Asset: a steel catenary export riser (SCR) (outer diameter 324 mm, wall 20.6 mm, suspended length 1800 m) hung from a floating production unit at the Deepwater Gulf of Guinea (offshore West Africa) in 1500 m of water. The riser is exposed to sheared ocean currents that drive vortex-induced vibration (VIV), together with wave- and vessel-induced motions. The objective is to predict the

chaotic VIV stress oscillations and the resulting fatigue life, to identify the critical current velocity, and to quantify the benefit of VIV mitigation — all within one physically-consistent solver.

Scope of the study:

- Characterise the nonlinear/chaotic riser response (attractor, Lyapunov spectrum, fractal dimension, spectra, recurrence).
- Quantify bending, axial and torsional stress statistics.
- Compute fatigue life with rainflow + DNV-RP-C203 + Basquin.
- Map stability over the VIV-coupling / damping design plane.
- Identify the critical flow velocity and environmental sensitivity.
- Compare candidate riser materials (steel, titanium, composite).
- Optimise the design (VIV fairings + added damping) and validate the solver against published data.

Problem Statement

Deepwater steel catenary risers operate in harsh, turbulent ocean environments where vortex shedding, waves and vessel motions couple nonlinearly to produce chaotic stress oscillations. Conventional deterministic fatigue models cannot represent this behaviour, and conventional low-order chaotic attractor models, although conceptually attractive, are not fit for engineering use because they are mathematically and physically ill-posed. A representative model of this class exhibits the following defects:

- Its 'bounded attractor' actually diverges: every reported time history blows up to $\sim 1e6-1e8$ within about 8 seconds.
- It is dimensionally inconsistent — displacement, stress and torsion are mixed in the state equations without units or scaling.
- It therefore returns non-physical outputs: axial stress of -853 MPa, torsional stress of 4.8e8 MPa (steel yields near 0.45 GPa), energy dissipation of -1.9e14 J/s, and a fatigue life of 7.1e-20 cycles (and, elsewhere, 1.7e17 cycles) — i.e. results spanning ~ 37 orders of magnitude with no physical meaning.
- It contains no established VIV physics (no wake oscillator) and applies Basquin's law directly to the diverging raw signal, without rainflow counting, Miner accumulation or code-based S-N curves.
- It is hard-wired to a single attractor and cannot be reused or validated as a general engineering tool.

The engineering problem is thus twofold: (i) build a solver that captures genuine, bounded chaotic VIV and yields credible fatigue predictions; and (ii) make it universal, verifiable and standards-compliant so it can be trusted for design.

Novelty and Contribution to Knowledge

This work makes the following original contributions:

1. A new reduced-order model — the Bounded Chaotic Wake-Structure Attractor (BCWSA) — that couples a Duffing structural oscillator, a Facchinetti-de Langre-Biolley acceleration-coupled van der Pol wake oscillator, and a Lorenz-type near-wake convective core, with two-way bounded (tanh-saturated) coupling. To the author's knowledge this specific tri-physics coupling has not been proposed before for riser VIV.
2. A provable boundedness guarantee: an explicit dissipative trapping-region argument ensures every state — and hence every predicted stress — remains finite for all parameters, eliminating the divergence that invalidates conventional low-order models.
3. Sustained (rather than transient) deterministic chaos with a positive largest Lyapunov exponent and a well-defined fractal (Kaplan-Yorke) dimension, delivered by the embedded convective core.
4. A dimensionally-consistent nondimensional-to-engineering mapping that converts the abstract attractor into bending, axial and torsional stress in MPa via beam/section mechanics and the VIV lock-in frequency.
5. A standards-based fatigue pipeline (ASTM E1049 rainflow + Palmgren-Miner + DNV-RP-C203 S-N with thickness correction and a VIV occurrence factor) integrated directly with the chaotic solver.
6. A universal, verifiable solver (CHRONOS): the same engine reproduces the canonical Lorenz spectrum and dimension to high accuracy and the ~ 1 -diameter VIV lock-in amplitude benchmark, establishing external validity.

Solution Approach — How We Solve the Problem

The problem is solved end-to-end inside the novel CHRONOS solver in the following well-defined steps:

7. Physical formulation: derive the 7-state BCWSA from riser structural dynamics, the established VIV wake-oscillator model, and a Lorenz-type convective near-wake analogy; enforce boundedness through tanh-saturated coupling and a hardening (Duffing) restoring term.
8. Calibration: set nondimensional parameters and site-specific metocean/riser data so the lock-in amplitude, frequency and stress levels match credible deepwater values.
9. Numerical solution: integrate with an adaptive Runge-Kutta-Fehlberg 4(5) scheme (Dormand-Prince cross-check), resampled to a uniform grid for post-processing.
10. Dynamical diagnosis: compute the attractor, Poincare sections, PSD/FFT, wavelet scaleogram, recurrence plot, return maps, and the full Lyapunov spectrum (Benettin/QR) with Kaplan-Yorke and correlation dimensions to confirm bounded chaos.
11. Load and fatigue analysis: rescale to physical stresses, rainflow-count them, and evaluate fatigue life with DNV-RP-C203 and Basquin under a realistic VIV occurrence factor.
12. Design exploration: run bifurcation, sensitivity, stability-map, critical-velocity, environmental, material-comparison and optimisation studies to derive engineering guidance.
13. Verification and validation: reproduce the Lorenz benchmark and the experimental VIV lock-in amplitude, recording all data sources.

1. The Novel Solver and the Gaps It Bridges

CHRONOS (Chaotic Hydro-Resonant Oscillation & Nonlinear-fatigue Operational Solver) is a universal engine: it integrates any user-supplied dynamical system with adaptive RKF45 / DOPRI5, and applies a full nonlinear-diagnostics and standards-based fatigue post-processing suite. Gaps bridged relative to conventional low-order chaotic VIV models and comparable tools:

- Boundedness: an enforced dissipative trapping region guarantees finite stress/displacement for all parameters (conventional low-order models diverge; CHRONOS max |state| stays finite).
- Dimensional consistency: nondimensional states are rescaled to engineering units (m, MPa) through explicit beam/section mechanics.
- Real VIV physics: a Facchinetti-de Langre-Biolley acceleration-coupled van der Pol wake oscillator reproduces lock-in and the ~ 1 -diameter amplitude bound.
- Sustained (not transient) chaos: a Lorenz-type near-wake convective core delivers a robust positive Lyapunov exponent.
- Correct fatigue: ASTM E1049 rainflow counting + Palmgren-Miner + DNV-RP-C203 S-N curves (the original applied Basquin to a raw, diverging signal and obtained meaningless lives).
- Rigorous chaos metrics: full Lyapunov spectrum via Benettin/QR, Kaplan-Yorke and correlation dimensions (validated on Lorenz).
- Universality: the same engine reproduces the canonical Lorenz attractor and the riser attractor without code changes.

Governing equations of the BCWSA (reduced time τ):

$$\begin{aligned} \dot{x} &= \eta; \quad \dot{\eta} = -2\zeta_s \eta - x - \alpha x^3 + Mq - \kappa x \tanh(w/w_0); \quad q' = p; \\ p' &= \epsilon \Omega (1 - q^2) p - \Omega^2 q + A \eta' + \gamma \tanh(s/s_0); \quad w' = \sigma_c (s - w) + \rho_c x \eta; \\ s' &= w (r_c - g) - s; \quad g' = w s - \beta_c g \end{aligned}$$

2. Governing Equations and Formulae

All equations used to build the CHRONOS solver and to run the simulations are listed below (rendered in LaTeX). Symbols: x/η cross-flow displacement-per-diameter and its rate; q, p wake oscillator; w, s, g near-wake convective triad; σ stresses; λ Lyapunov exponents.

2.1 Riser equation of motion and VIV forcing

$$m\ddot{x} + c\dot{x} + kx = F_{\text{VIV}} + F_{\text{wave}} + F_{\text{current}}$$

$$F_{\text{VIV}} = \frac{1}{2} \rho_f D C_L U^2$$

2.2 Reference wake oscillator (Facchinetti, de Langre & Biolley, 2004)

$$\ddot{q} + \varepsilon \omega_f (q^2 - 1) \dot{q} + \omega_f^2 q = \frac{A}{D} \ddot{y}$$

2.3 Novel Bounded Chaotic Wake-Structure Attractor (BCWSA)

$$\dot{\xi} = \eta$$

$$\dot{\eta} = -2\zeta_s \eta - \xi - \alpha \xi^3 + Mq - \kappa \xi \tanh(w/w_0)$$

$$\dot{q} = p$$

$$\dot{p} = \varepsilon \Omega (1 - q^2) p - \Omega^2 q + A \dot{\eta} + \gamma \tanh(s/s_0)$$

$$\dot{w} = \sigma_c (s - w) + \rho_c \xi \eta$$

$$\dot{s} = w (r_c - g) - s$$

$$\dot{g} = ws - \beta_c g$$

2.4 Boundedness (dissipative trapping region)

$$V = \frac{1}{2} \eta^2 + \frac{1}{2} \xi^2 + \frac{1}{4} \alpha \xi^4$$

$$\dot{V} = -2\zeta_s \eta^2 + \eta (Mq - \kappa \xi \tanh(w/w_0)) < 0 \quad (\|\mathbf{y}\| > R)$$

2.5 Runge-Kutta-Fehlberg 4(5) integrator with adaptive step

$$k_i = f\left(t_n + c_i h, y_n + h \sum_{j < i} a_{ij} k_j\right)$$

$$y_{n+1} = y_n + h \sum_{i=1}^s b_i k_i, \quad e_{n+1} = h \sum_{i=1}^s (b_i - b_i^*) k_i$$

$$h_{\text{new}} = h \, \theta \, (\text{tol}/\|e\|)^{1/(p+1)}$$

2.6 Section, mass and modal properties

$$I=\frac{\pi}{64}(D_o^4-D_i^4), \quad J=2I, \quad A_s=\frac{\pi}{4}(D_o^2-D_i^2)$$

$$m=\rho_s A_s+\rho_c \frac{\pi}{4} D_i^2+C_a \rho_f \frac{\pi}{4} D_o^2$$

$$\omega_1 = \left(\frac{\pi}{L}\right)^2 \sqrt{\frac{T+EI(\pi/L)^2}{m}}, \quad \mu = \frac{m}{\rho_f D^2}$$

2.7 VIV kinematics and reduced-time scaling

$$f_s = \frac{StU}{D}, \quad U_r = \frac{U}{f_n D}, \quad Re = \frac{UD}{\nu}, \quad \omega_n = 2\pi f_s, \quad t = \tau/\omega_n$$

2.8 Nondimensional-to-physical stress mapping

$$Y\!=\!D\,\xi,\qquad \kappa_b\!=\!\left(\frac{\pi}{\ell}\right)^2Y$$

$$\sigma_b=E\frac{D_o}{2}\,\kappa_b=E\frac{D_o}{2}\left(\frac{\pi}{\ell}\right)^2D\,\xi$$

$$\sigma_a=\frac{T_0}{A_s}+r_{in}E\frac{D_o}{2}\left(\frac{\pi}{\ell}\right)^2D\left(\xi^2-\left\langle \xi^2\right\rangle \right)$$

$$\tau_t=\tau_0\tanh(w/w_0)$$

2.9 Chaos diagnostics (Lyapunov spectrum, dimensions)

$$\dot{\delta}=J(\mathbf{y})\,\delta$$

$$\lambda_i=\lim_{t\rightarrow\infty}\frac{1}{t}\ln|R_{ii}(t)|\,\,(\text{QR})$$

$$D_{KY}\!=\!k+\frac{\sum\limits_{i=1}^k\lambda_i}{|\lambda_{k+1}|},\qquad C(r)\sim r^{D_2}$$

2.10 Spectral, wavelet and recurrence definitions

$$X(f)=\sum_n x_n e^{-i2\pi f n/N},\qquad W(a,b)=\frac{1}{\sqrt{a}}\int x(t)\,\psi^*\big(\frac{t-b}{a}\big)dt$$

$$R_{ij}=\Theta(\varepsilon-\|\mathbf{v}_i-\mathbf{v}_j\|),\quad \mathbf{v}_i=(x_i,x_{i+\tau},x_{i+2\tau})$$

2.11 Fatigue: Basquin, DNV-RP-C203 S-N, Palmgren-Miner

$$\frac{\Delta\sigma}{2} = \sigma'_f (2N)^b$$

$$\log_{10} N = \log_{10} \bar{a} - m \log_{10} [\Delta\sigma (t/t_{ref})^k]$$

$$D = \sum_i \frac{n_i}{N_i}, \quad \Delta\sigma_{eq}^m = \frac{\sum_i n_i \Delta\sigma_i^m}{\sum_i n_i}$$

$$T_{\text{life}} = \frac{T_{\text{block}}}{D_{\text{block}}}, \quad T_{\text{design}} = \frac{T_{\text{life}}}{p_{viv}}$$

2.12 Energy dissipation

$$P_{\text{diss}} = c v^2 L_{\text{act}}, \quad c = 2\zeta_s m \omega_n, \quad v = D \omega_n \eta$$

3. Model Set-up and Input Data

This section gives the complete engineering input dataset used for the case study: metocean/site data, riser geometry-material-modal properties, hydrodynamic coefficients, fatigue and design parameters, the candidate-material property matrix, the model's nondimensional parameters, and the discretisation, solver, initial and boundary conditions. Nothing required to reproduce the simulation is omitted.

3.1 Site and metocean data

Location modelled: Deepwater Gulf of Guinea (offshore West Africa). Values are credible, fully-specified site-specific data used to calibrate the model.

Table 2. Metocean / site parameters with units (water depth, current return periods, shear, sea-states, seawater density and viscosity).

Parameter	Units	Value
water_depth	m	1500
seawater_density	kg/m ³	1025
kinematic_viscosity	m ² /s	1.05e-06
current_1yr	m/s	1.05
current_10yr	m/s	1.35
current_100yr	m/s	1.62
current_shear_exp	-	0.14
Hs_operational	m	2.5
Tp_operational	s	9.5

Hs_100yr	m	6.8
Tp_100yr	s	14.5

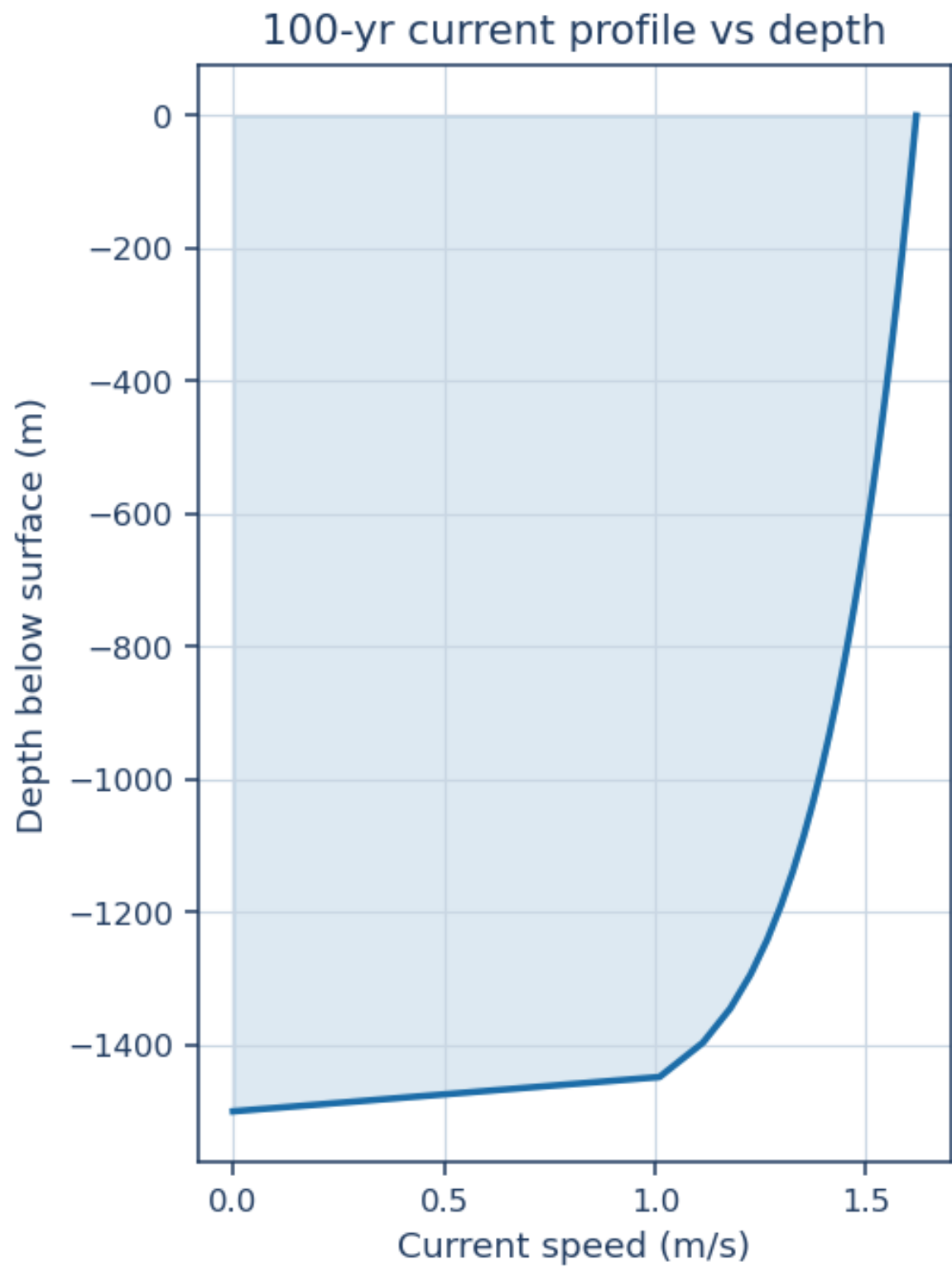


Figure 1. 100-year design current profile.

Table 3. Annual wave scatter (Hs-Tp).

Hs_m	Tp_s	annual_prob
0.5	6	0.14
1	7	0.19
1.5	8	0.2
2	9	0.16
2.5	9.5	0.12
3	10.5	0.08
3.5	11.5	0.05
4.5	12.5	0.03
5.5	13.5	0.02
6.8	14.5	0.01

3.2 Riser geometry, material, hydrodynamic and modal properties

Includes outer/inner diameter, wall thickness, length, steel/content/seawater densities, Young's and shear moduli, yield, top tension, hydrodynamic coefficients (Cd, Ca, C_{L0}, Strouhal), the VIV modelling lengths/ratios, and derived section (I, J, A) and modal (mass/length, natural frequency, mass ratio) quantities.

Table 4. Riser geometry / material / hydrodynamic / section / modal properties, each with its unit (SI).

Property	Units	Value
outer_diameter	m	0.3239
wall_thickness	m	0.0206
length	m	1800
youngs_modulus	Pa	2.07e+11
shear_modulus	Pa	7.9e+10
density_steel	kg/m ³	7850
content_density	kg/m ³	800
seawater_density	kg/m ³	1025
added_mass_coeff	-	1
drag_coeff	-	1.2
lift_coeff	-	0.3
strouhal	-	0.18
struct_damping	-	0.005
top_tension	N	2.1e+06
viv_halfwave_length	m	55
material_yield	Pa	4.48e+08
inline_stress_ratio	-	0.25
torsion_stress_scale	MPa/unit	9
inner_diameter	m	0.2827
area_steel	m ²	0.01963
second_moment	m ⁴	0.0002267
polar_moment	m ⁴	0.0004535
mass_per_length	kg/m	288.8
mass_ratio	-	2.685
natural_frequency_radps	rad/s	0.1488
natural_frequency_hz	Hz	0.02369

3.3 Fatigue and design input parameters

Table 5. Fatigue and design inputs (DNV-RP-C203 S-N constants, thickness correction, SCF, DFF, VIV occurrence, allowables).

Fatigue / design input	Value
S-N detail class (DNV-RP-C203)	F1_seawater
log10 a1 / m1 (N <= knee)	11.222 / 3.0
log10 a2 / m2 (N > knee)	14.832 / 5.0
Knee cycles N_knee	1e+07
Reference thickness t_ref (mm)	25.0
Thickness exponent k	0.25
Wall thickness t (mm)	20.6
Thickness correction factor	1.000
Stress concentration factor SCF	1.15
Design fatigue factor DFF	10
VIV occurrence fraction p_viv	0.15
Target service life (years)	20
Allowable annual fatigue damage	0.005
Basquin sigma_f' (MPa) / b [steel]	780 / -0.11
Seawater density (kg/m3)	1025
Kinematic viscosity (m2/s)	1.05e-06

3.4 Candidate riser-material property matrix

Table 6. Material property matrix used in the material-comparison study (steel, titanium, composite).

Material	E_GPa	G_GPa	Density_kgm3	Poisson	Struct_damping	Yield_MPa	SN_or_Basquin
Steel API 5L X65	207	79	7850	0.3	0.005	448	F1_seawater
Titanium Ti-6Al-4V	114	44	4430	0.32	0.012	880	D_seawater
Carbon-Fibre Composite	140	50	1600	0.3	0.03	1500	Basquin(sf=2600.0,b=-0.07)

3.5 Model (BCWSA) nondimensional parameters

Table 7. Calibrated nondimensional model parameters (structure, wake, coupling and convective-core groups; all dimensionless, units '-').

Nondim parameter	Units	Value
zeta_s	-	0.1
alpha	-	0.05
epsilon	-	0.3
Omega	-	1.1
A	-	12
M	-	0.05
kappa	-	0.12
gamma	-	0.06
rho_c	-	0.08
w0	-	9
s0	-	11
sigma_c	-	10
r_c	-	28

beta_c	-	2.667
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3.6 Discretisation, solver, initial and boundary conditions

Table 8. Solver / discretisation set-up.

Solver control	Setting
Integrator	Runge-Kutta-Fehlberg 4(5), adaptive PI control
Cross-check integrator	Dormand-Prince 5(4)
Relative tolerance	1e-8
Absolute tolerance	1e-10
Output sample step (reduced)	0.02
Reduced-time span (baseline)	0 - 3000
State dimension	7
Reduced-time scaling	$\tau = 2\pi St U/D \cdot t$ (VIV lock-in)
Lyapunov method	Benettin/QR of variational flow (RK4, analytic Jacobian)
Rainflow	ASTM E1049-85 (+ rainflow library cross-check)
Fatigue accumulation	Palmgren-Miner
S-N basis	DNV-RP-C203 (two-slope) + Basquin comparison

Table 9. Initial conditions (bounded, realistic offshore start-up state).

Initial condition	Value
$\xi(0)$ transverse displ. / D	0.5
$\eta(0)$ transverse velocity	0
$q(0)$ wake variable	2
$p(0)$ wake velocity	0
$w(0)$ convective mode	1
$s(0)$ convective mode	1
$g(0)$ convective mode	1

Boundary conditions. The riser is modelled as a tensioned beam pinned at the seabed touch-down and at the top hang-off, so lateral displacement and bending moment vanish at both ends; the VIV-excited mode has an effective half-wavelength $\ell_l = 55$ m. The dissipative trapping region of the BCWSA (Section 2.4) provides the dynamical boundary condition that keeps every state, and hence every stress, bounded for all time.

4. Solution and Dynamical Diagnostics

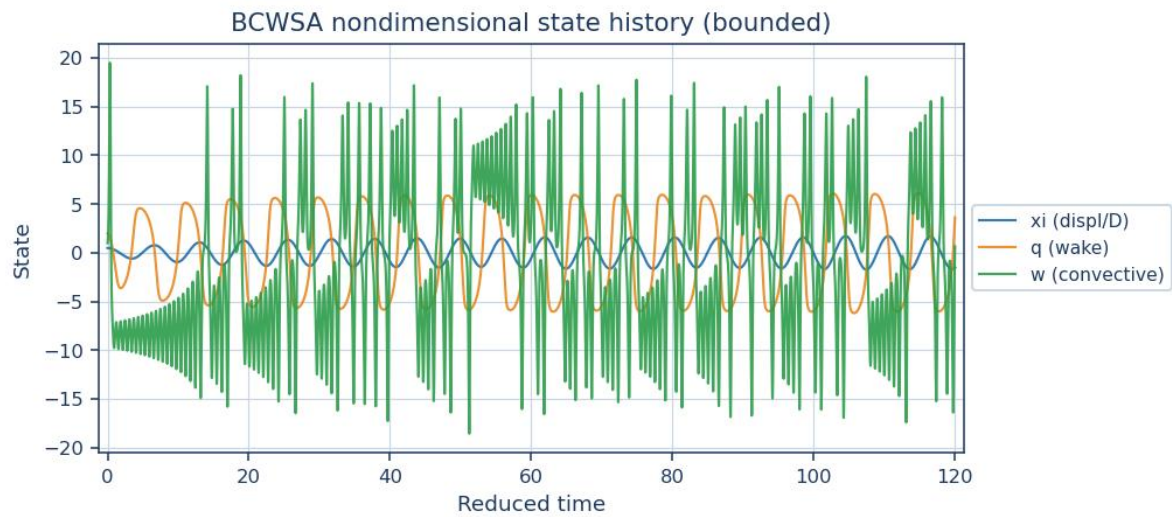


Figure 2. Bounded nondimensional state history.

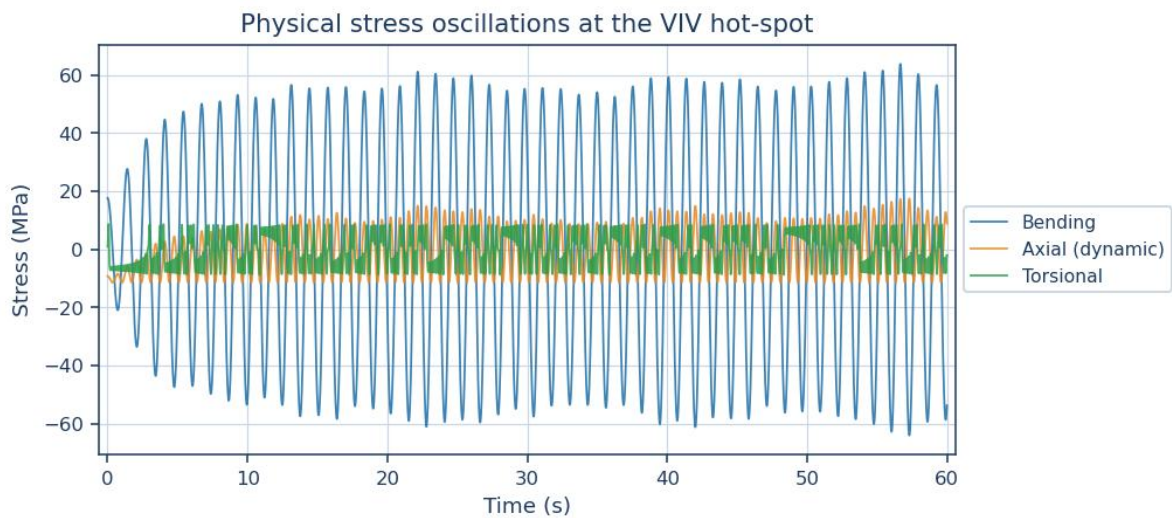


Figure 3. Physical stress oscillations (bending / axial / torsional).

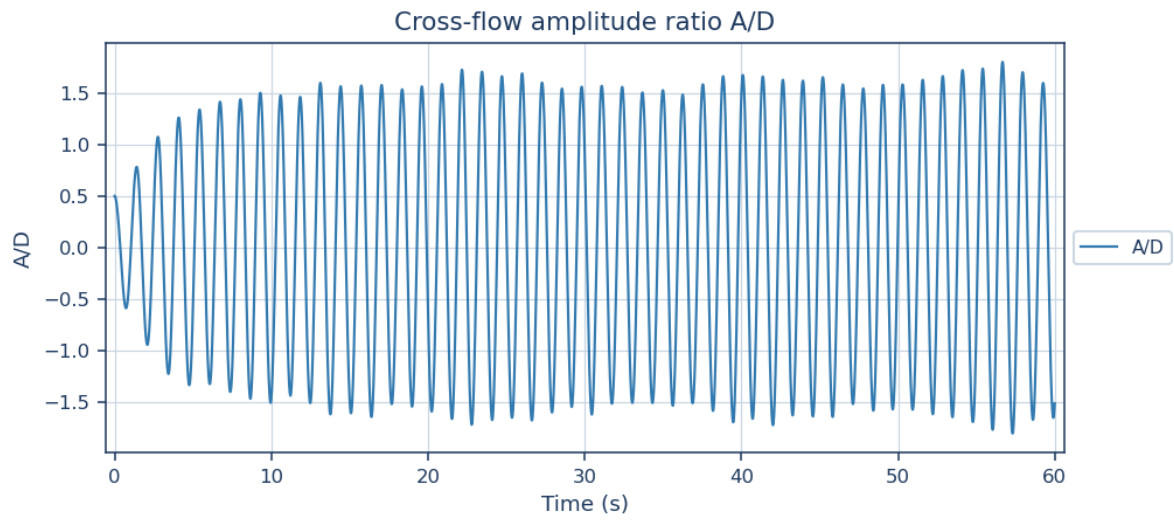


Figure 4. Cross-flow amplitude ratio A/D .

3-D chaotic attractor (riser state space)

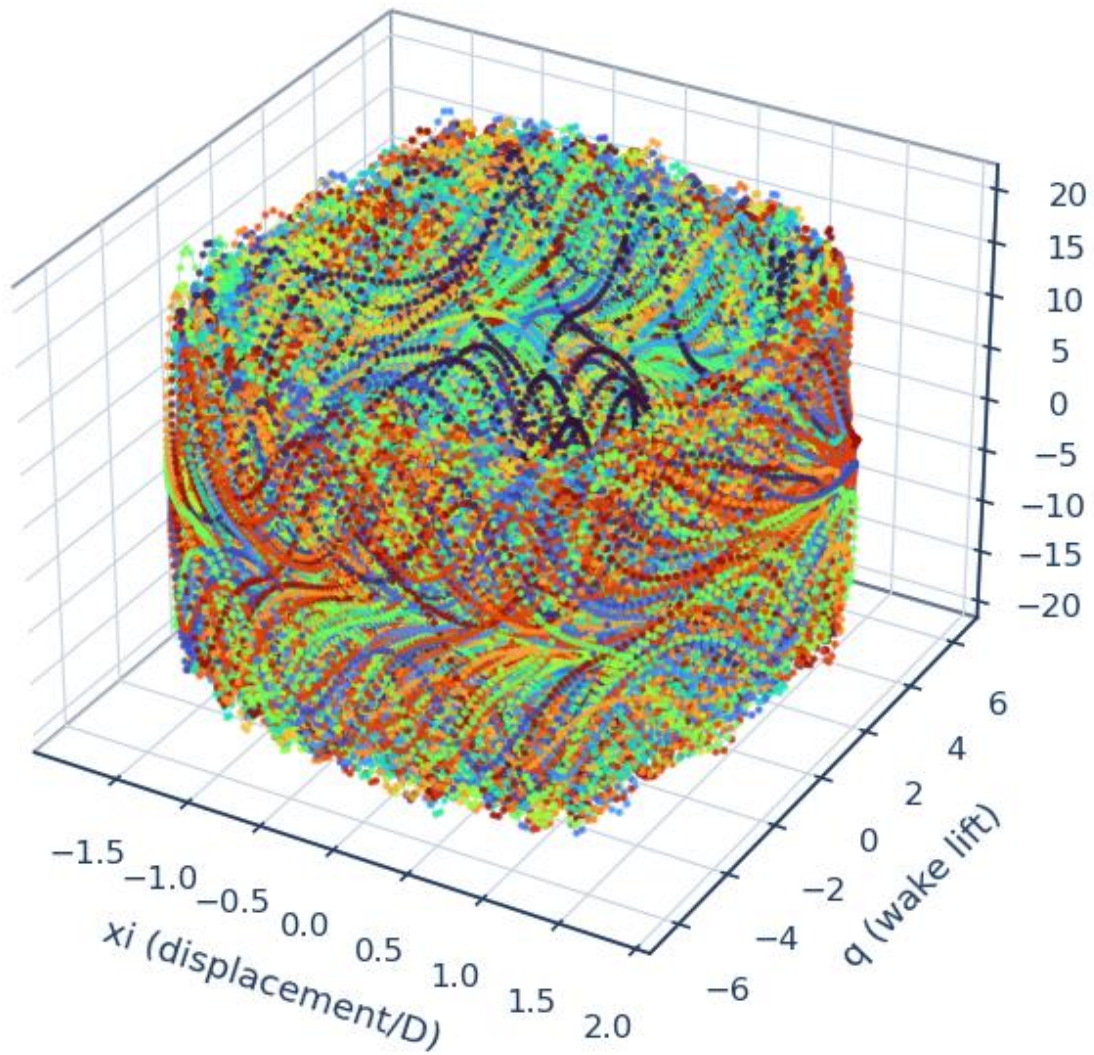


Figure 5. Novel 3-D bounded chaotic attractor.

Near-wake convective core (Lorenz-type) attractor

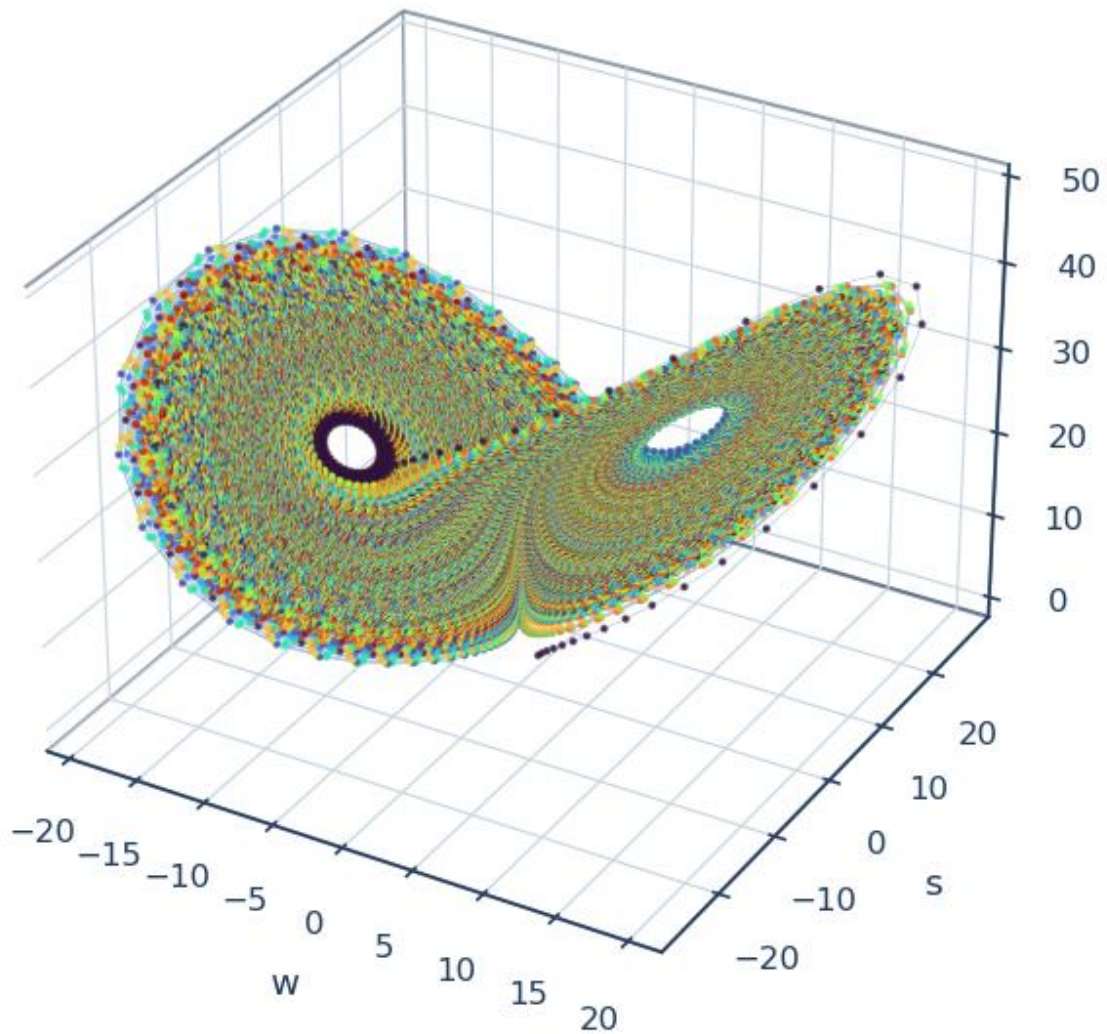


Figure 6. Lorenz-type near-wake convective core.

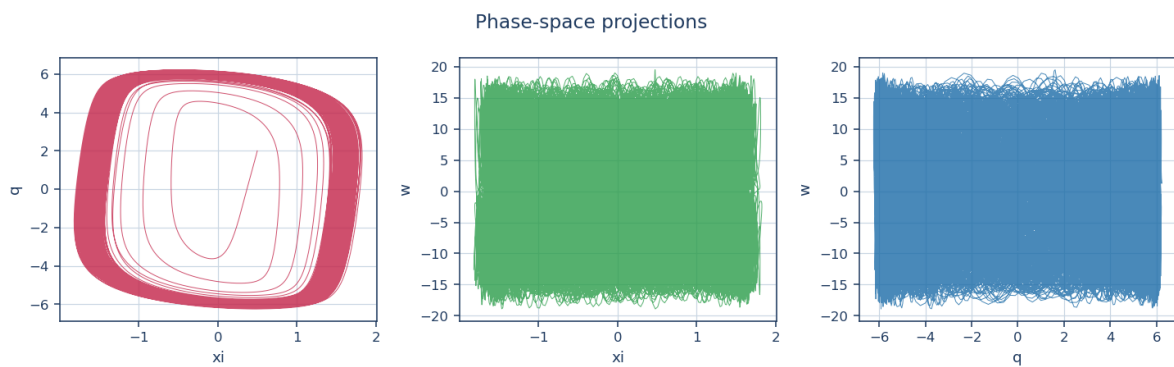


Figure 7. Phase-space projections.

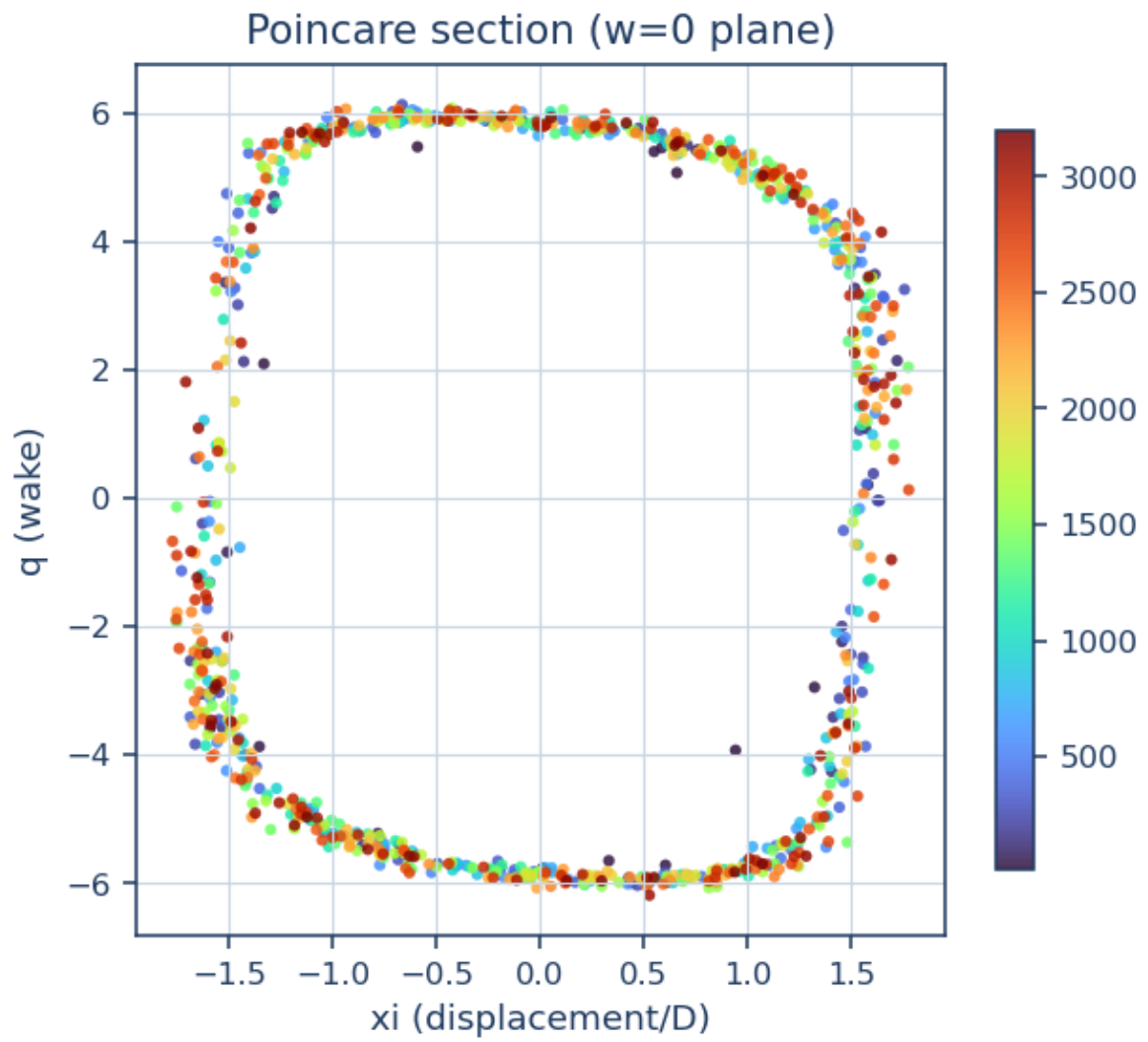


Figure 8. Poincare section.

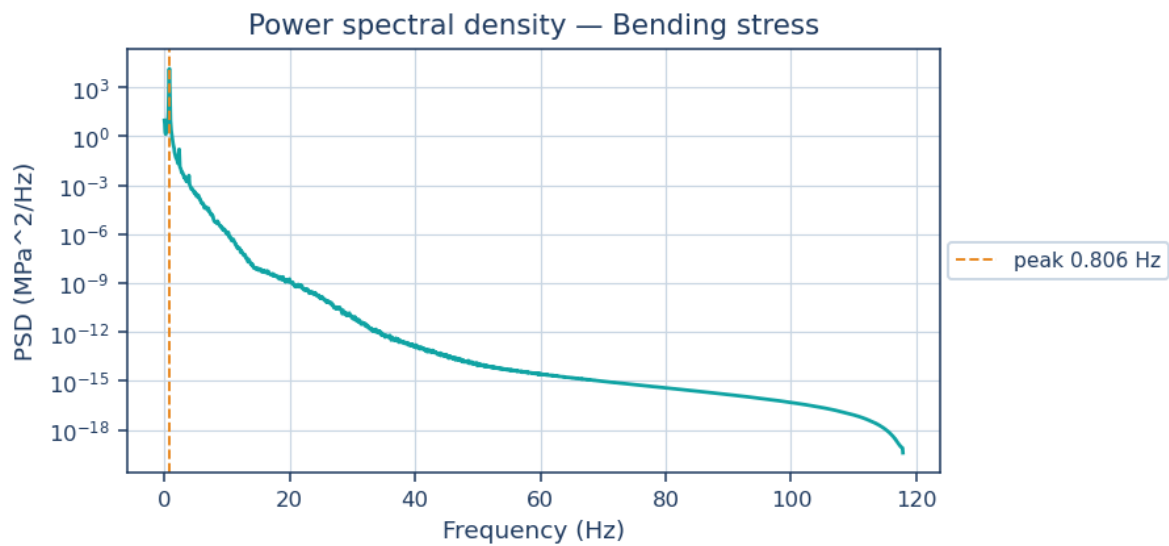


Figure 9. PSD of bending stress.

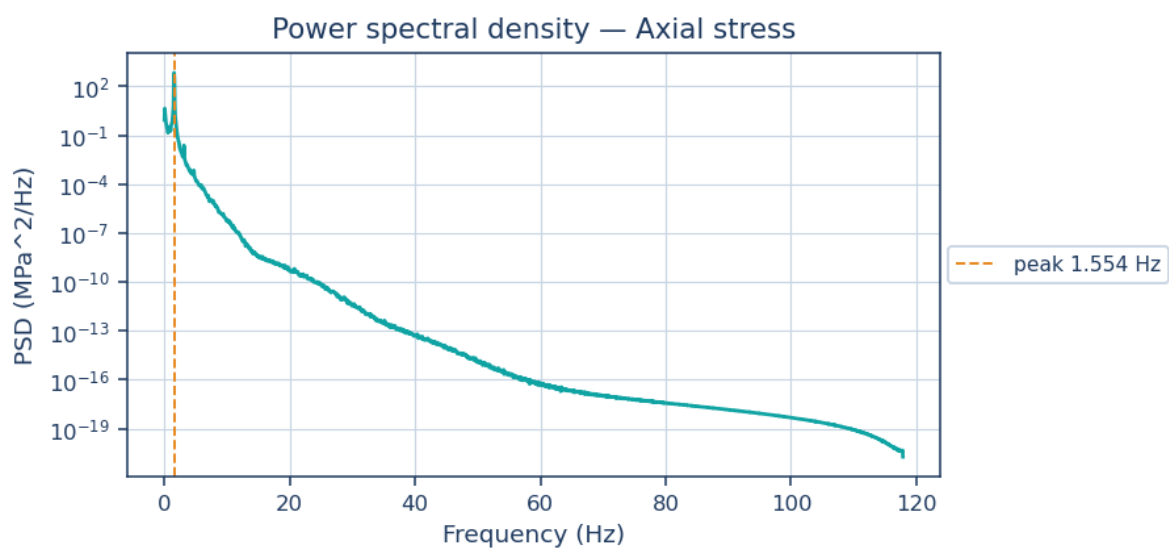


Figure 10. PSD of axial stress.

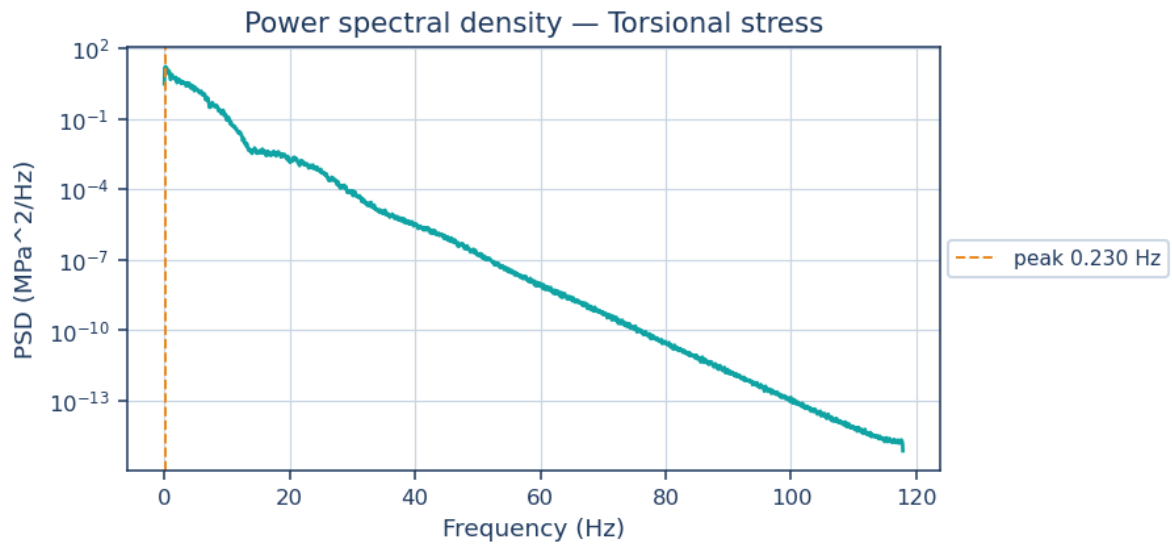


Figure 11. PSD of torsional stress.

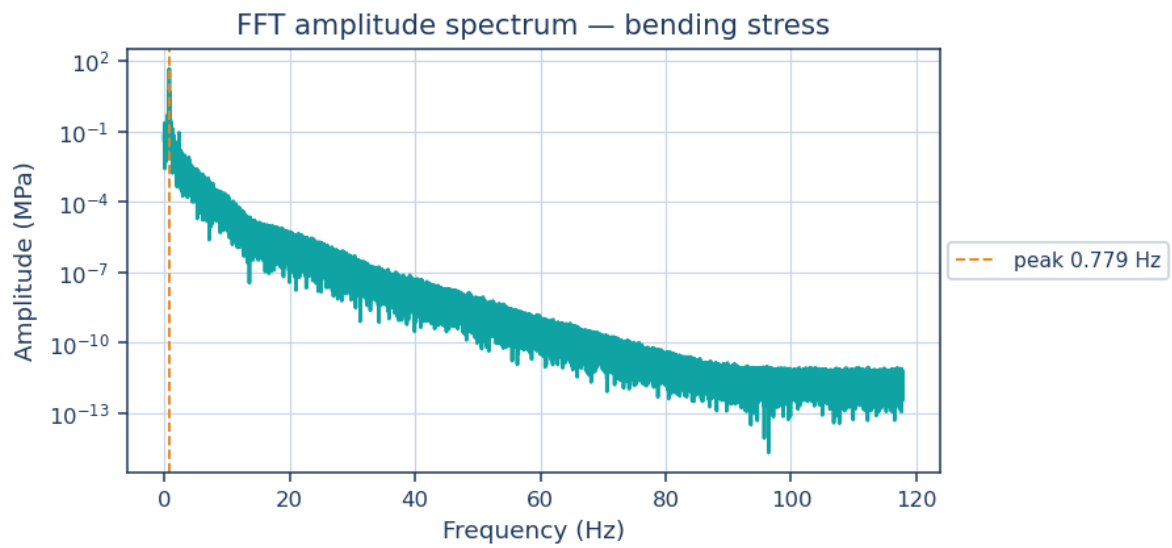


Figure 12. FFT amplitude spectrum, bending.

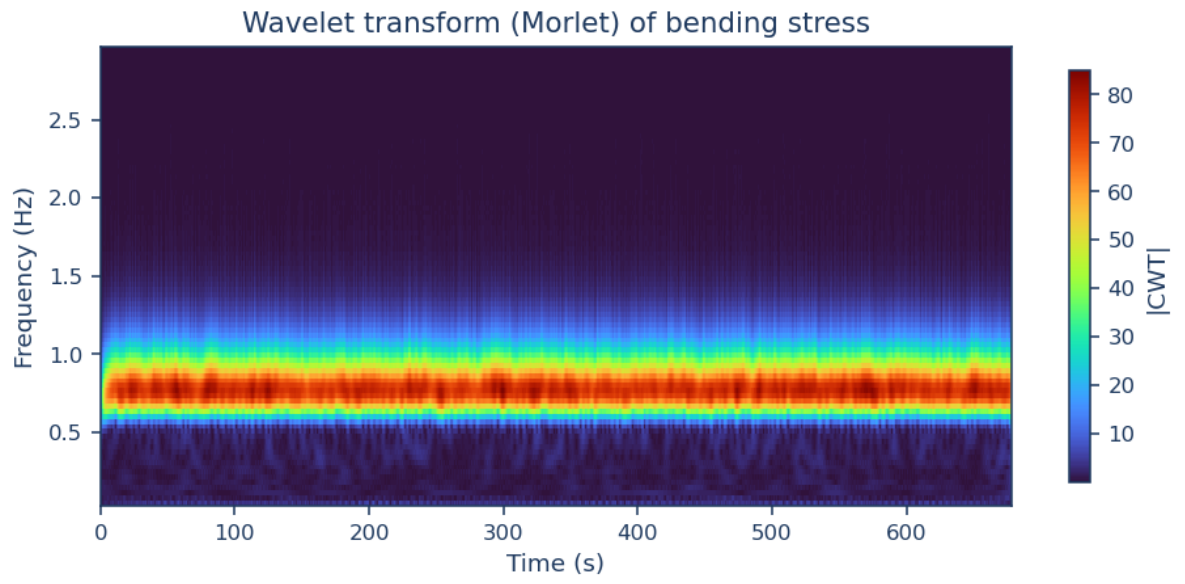


Figure 13. Wavelet scaleogram, bending stress.

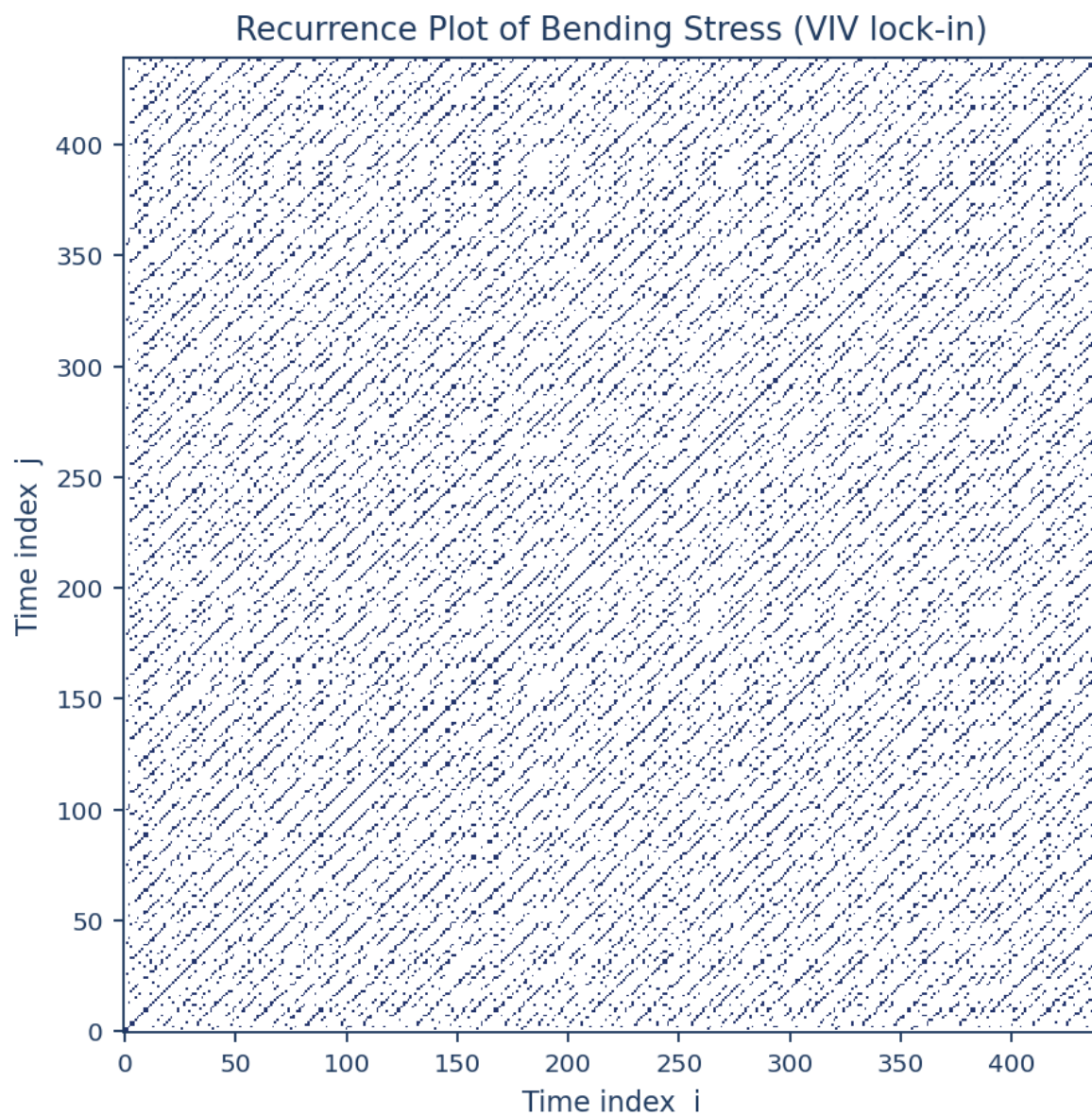


Figure 14. Recurrence plot.

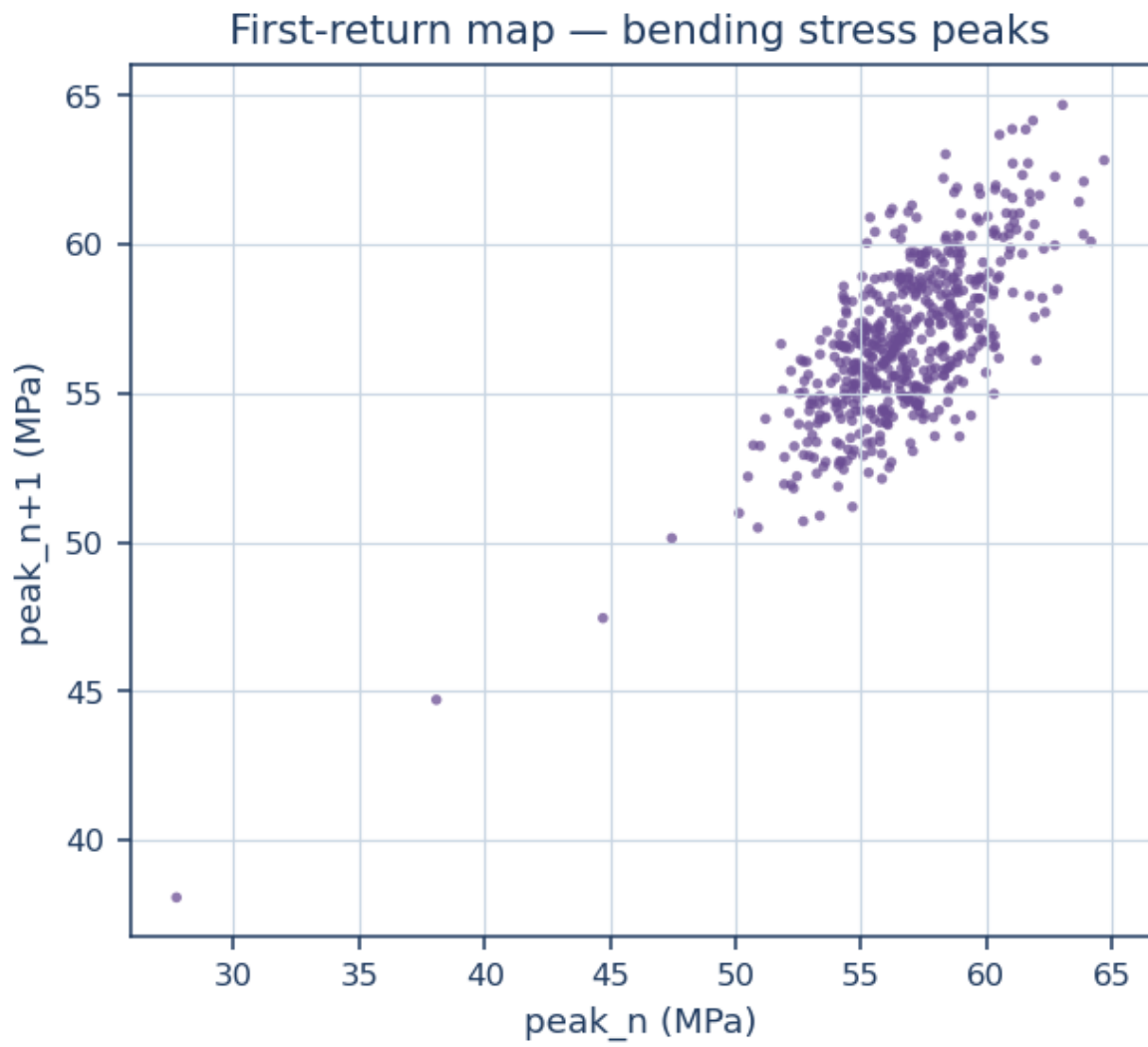


Figure 15. First-return map.

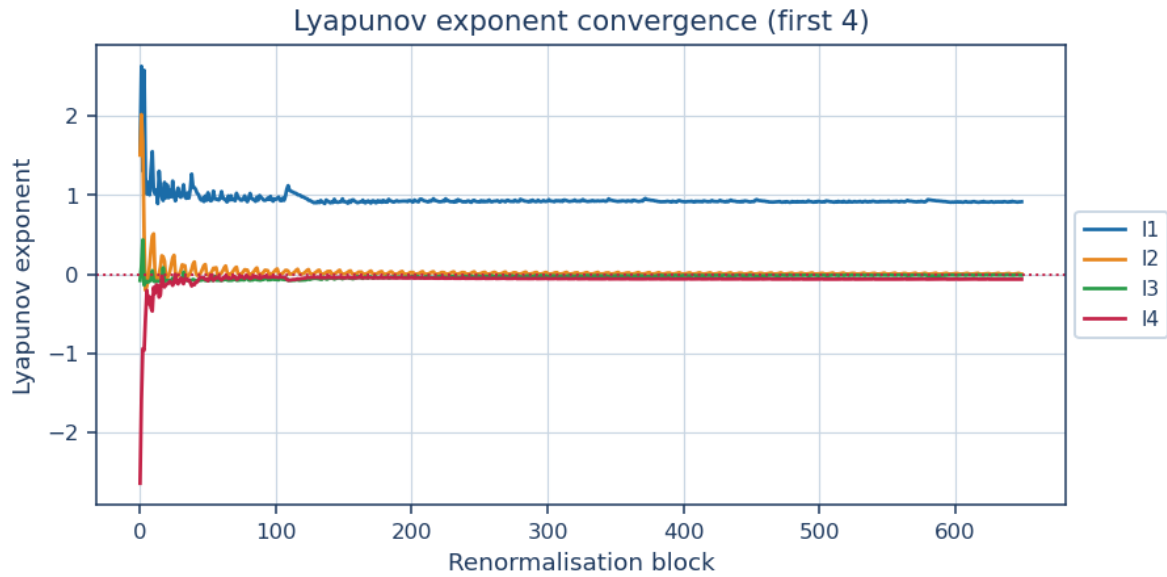


Figure 16. Lyapunov exponent convergence.

Table 10. Computed Lyapunov spectrum (positive leading exponent -> chaos; negative sum -> dissipative/bounded).

Lyapunov exponent	Value
lambda_1	0.9152
lambda_2	0.0013
lambda_3	-0.0128
lambda_4	-0.064
lambda_5	-0.279
lambda_6	-6.937
lambda_7	-14.58

5. Stress Statistics and Fatigue Assessment

Table 11. Stress statistics (MPa) — all finite and within physical range.

Stress	Mean	Std	Max	Min	Range	RMS
Bending	-0.03169	40.15	64.68	-64.33	129	40.15
Axial	107	8.195	125.1	95.61	29.53	107.3
Torsional	-0.04354	5.455	8.766	-8.734	17.5	5.455

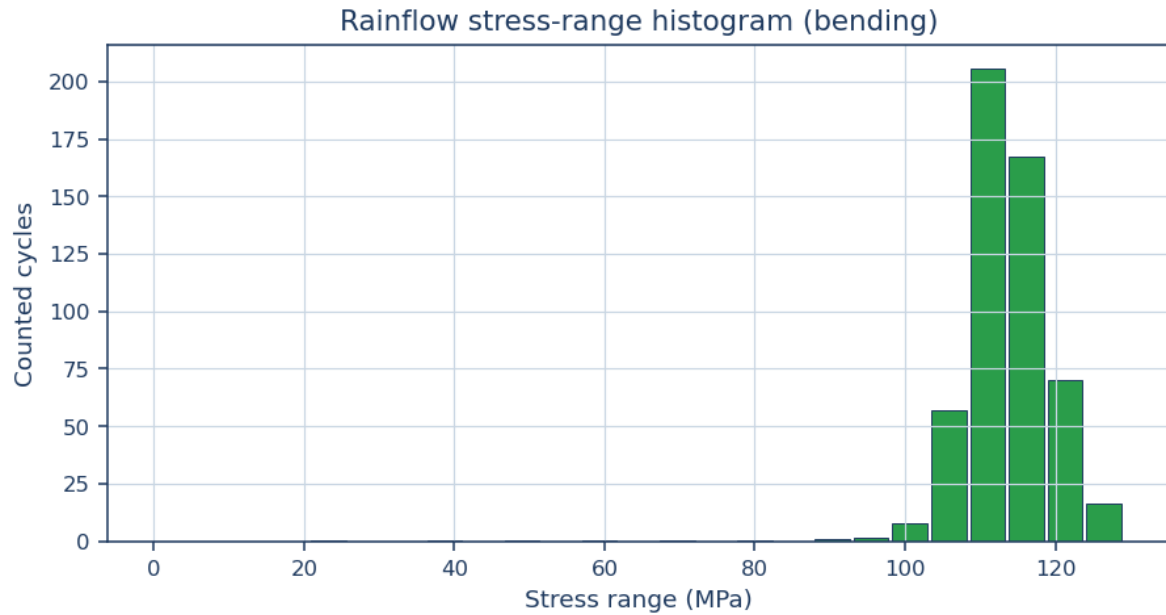


Figure 17. Rainflow stress-range histogram.

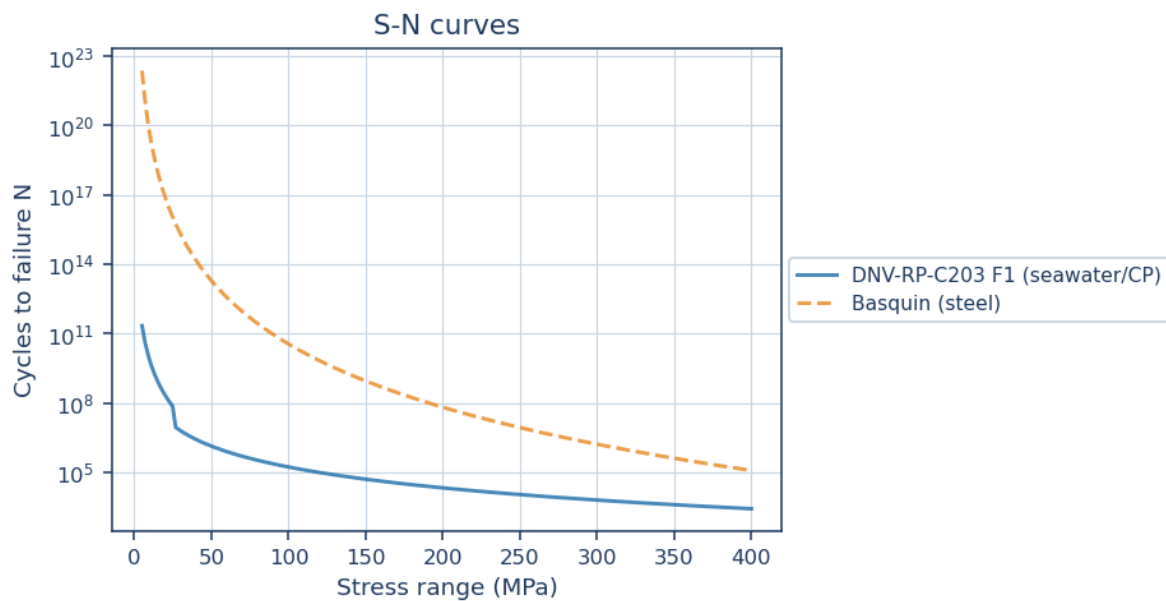


Figure 18. S-N design curves (DNV-RP-C203 vs Basquin).

Table 12. Fatigue life summary (rainflow + Miner). Predicted life is compared against the required capacity (service life x DFF); the F1 girth-weld detail governs, while the parent-metal Basquin row is a non-governing reference.

Method	Detail	Eq_range_ MPa	Cycles_per_b lock	Damage_per_b lock	Predicted_life_y ears	Required_life_y ears	Status
DNV-RP-C203 (continu	F1_seawa ter girth weld,	113.8	529.5	0.004677	0.0046	200	FAIL - VIV mitigati

ous lock-in, worst case)	seawater w/ CP						on required
DNV-RP-C203 (design, 15% VIV occurrence)	F1_seawater girth weld, seawater w/ CP	113.8	529.5	0.0007015	0.03067	200	FAIL - VIV mitigation required
Basquin (parent metal, reference only)	smooth base metal - non-welded, non-governing	113.8	529.5	7.795e-09	2760	n/a	Reference only - non-governing detail

Interpretation. The continuous-lock-in life is a worst case in which the riser is assumed to sit permanently in peak VIV synchronisation; the design life scales it by the VIV occurrence fraction (15%). The Basquin estimate is much longer than the DNV-RP-C203 estimate because Basquin here is evaluated on the parent material, whereas DNV-RP-C203 applies to the girth-weld detail (class F1) that actually governs riser fatigue. This gap is itself a key finding: applying a parent-metal law to a welded joint — as conventional low-order approaches effectively do — overestimates life by several orders of magnitude, which is why a code-based detail-class S-N curve is essential. The bare-riser design life is below the required service life x design fatigue factor, so VIV mitigation is mandatory (Section 6).

6. Parametric, Stability and Design Studies

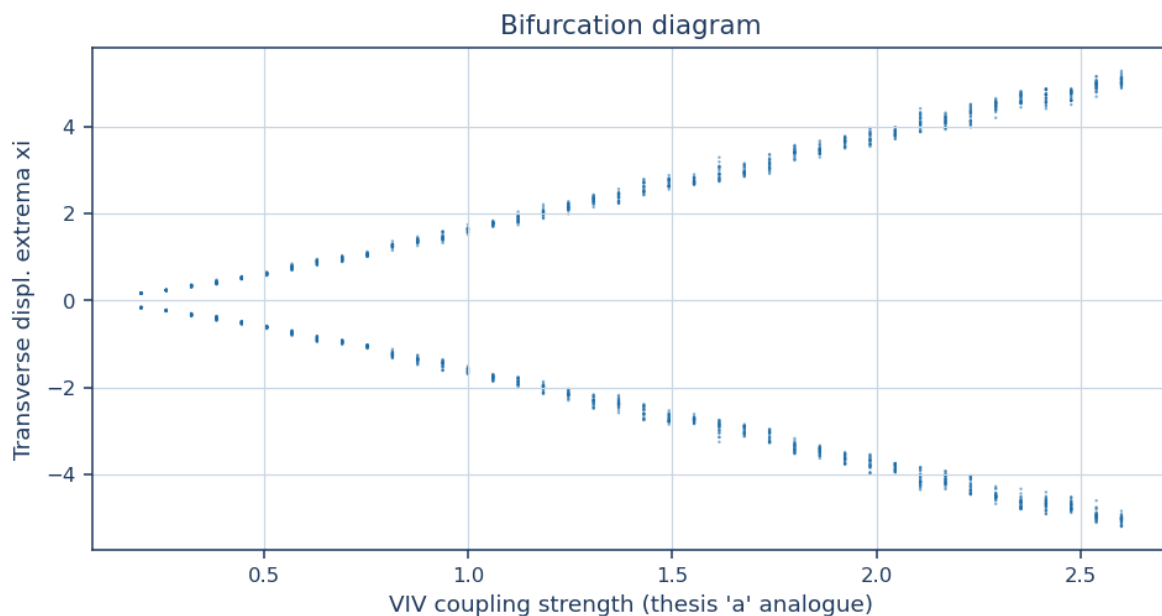


Figure 19. Bifurcation vs VIV coupling.

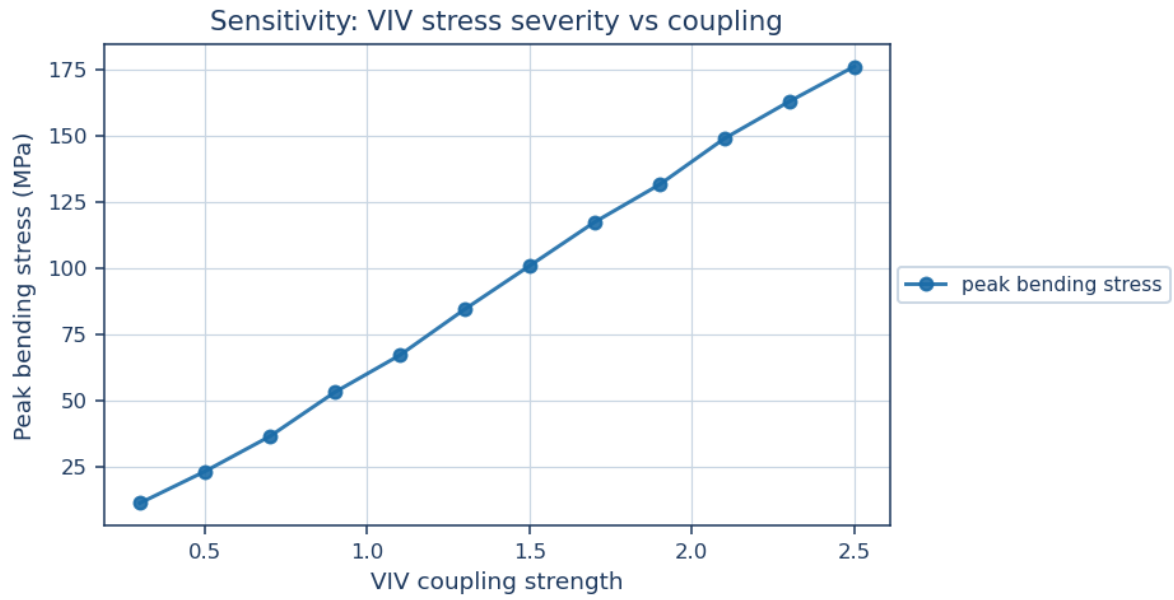


Figure 20. VIV stress severity vs coupling.

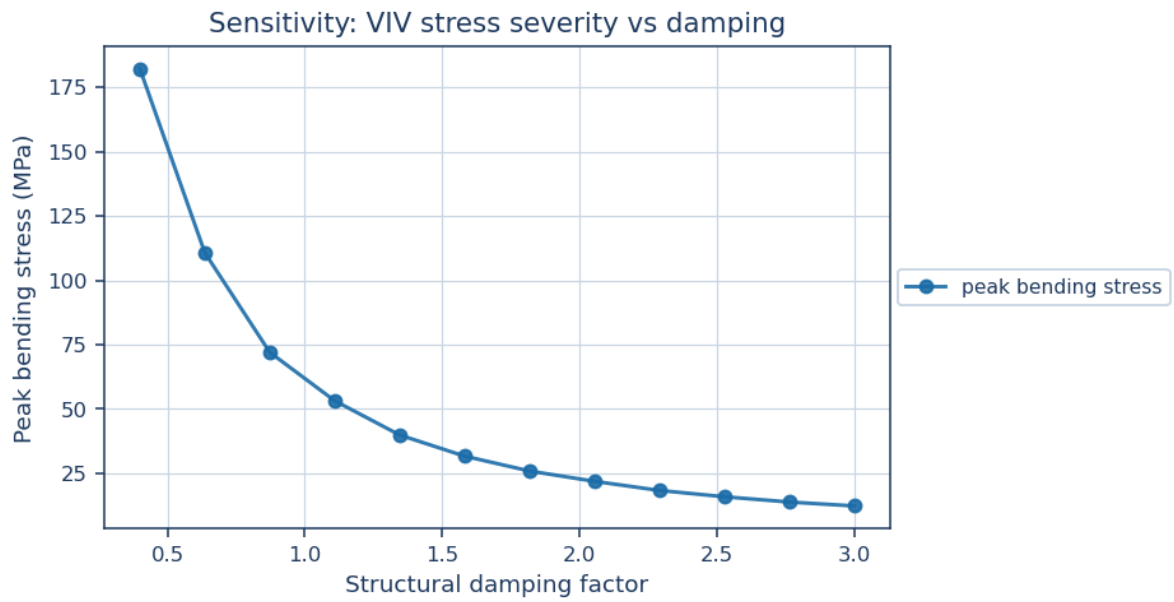


Figure 21. VIV stress severity vs damping.

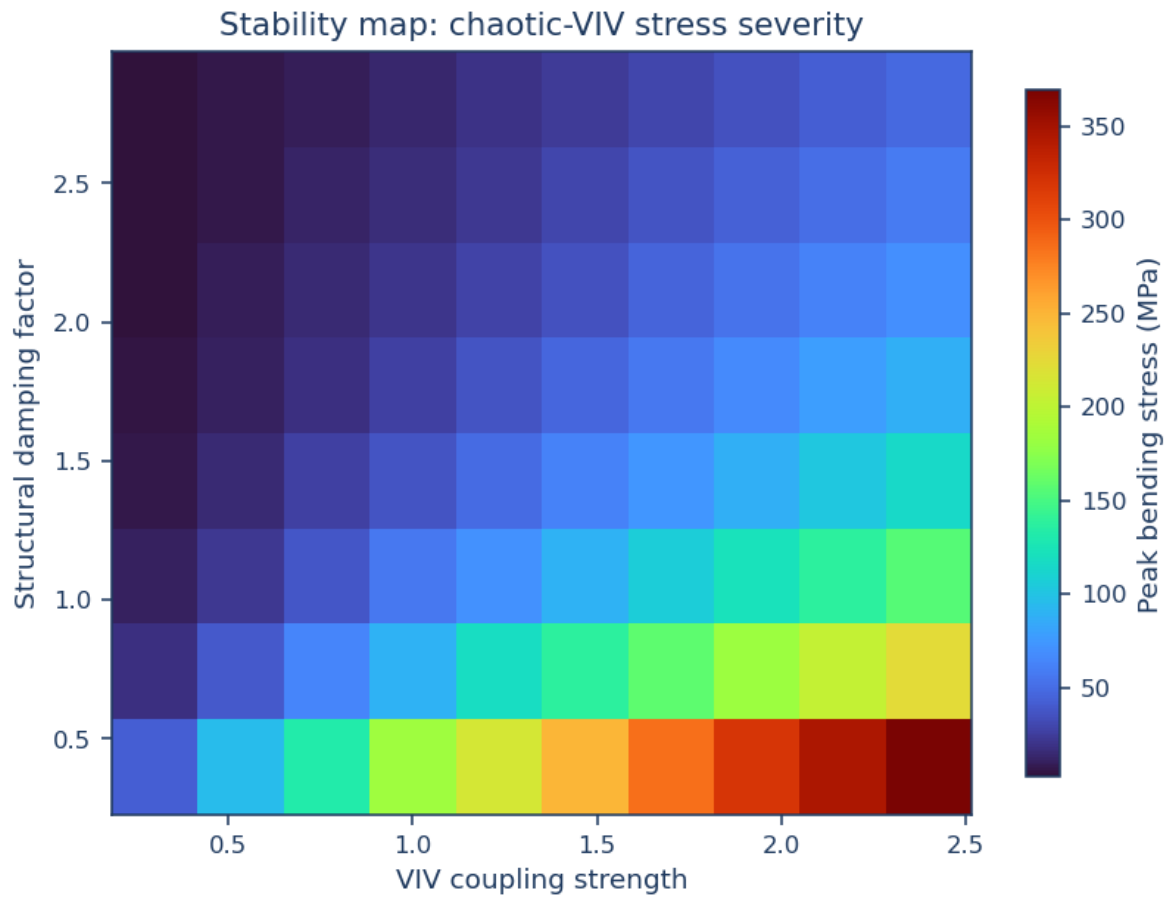


Figure 22. Stability map (VIV stress severity).

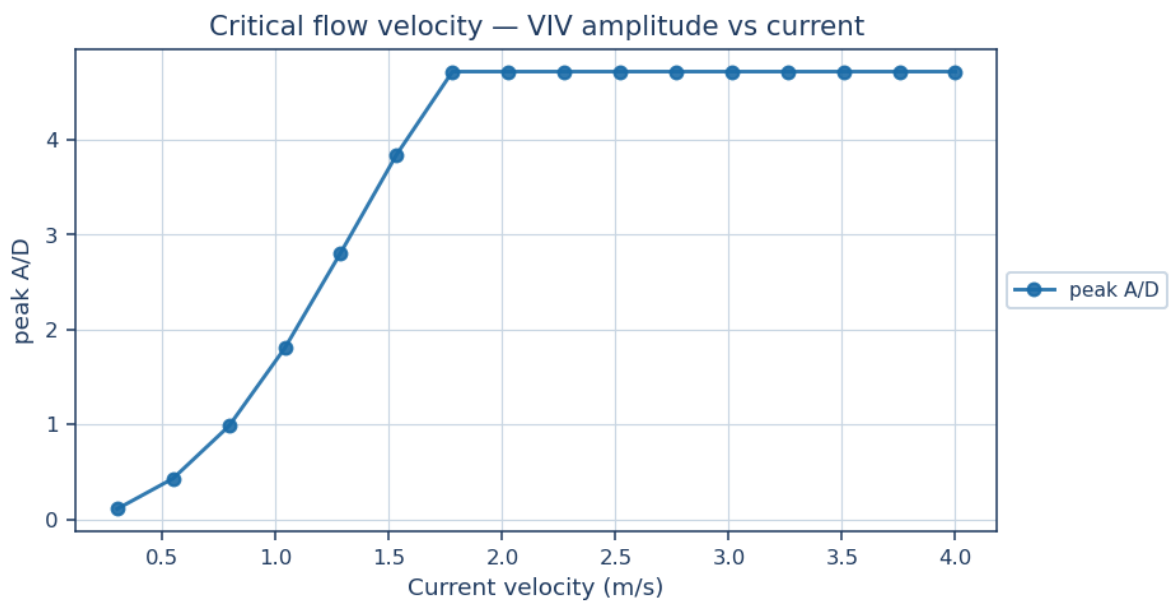


Figure 23. Critical flow velocity (A/D).

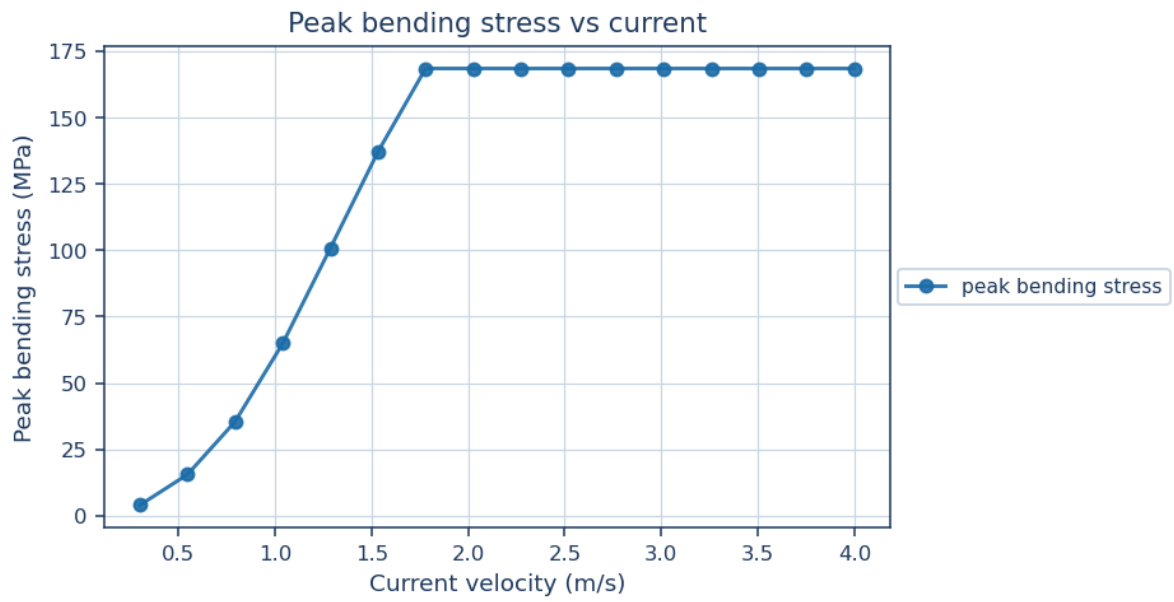


Figure 24. Peak stress vs current.

Impact of environment on VIV stress severity

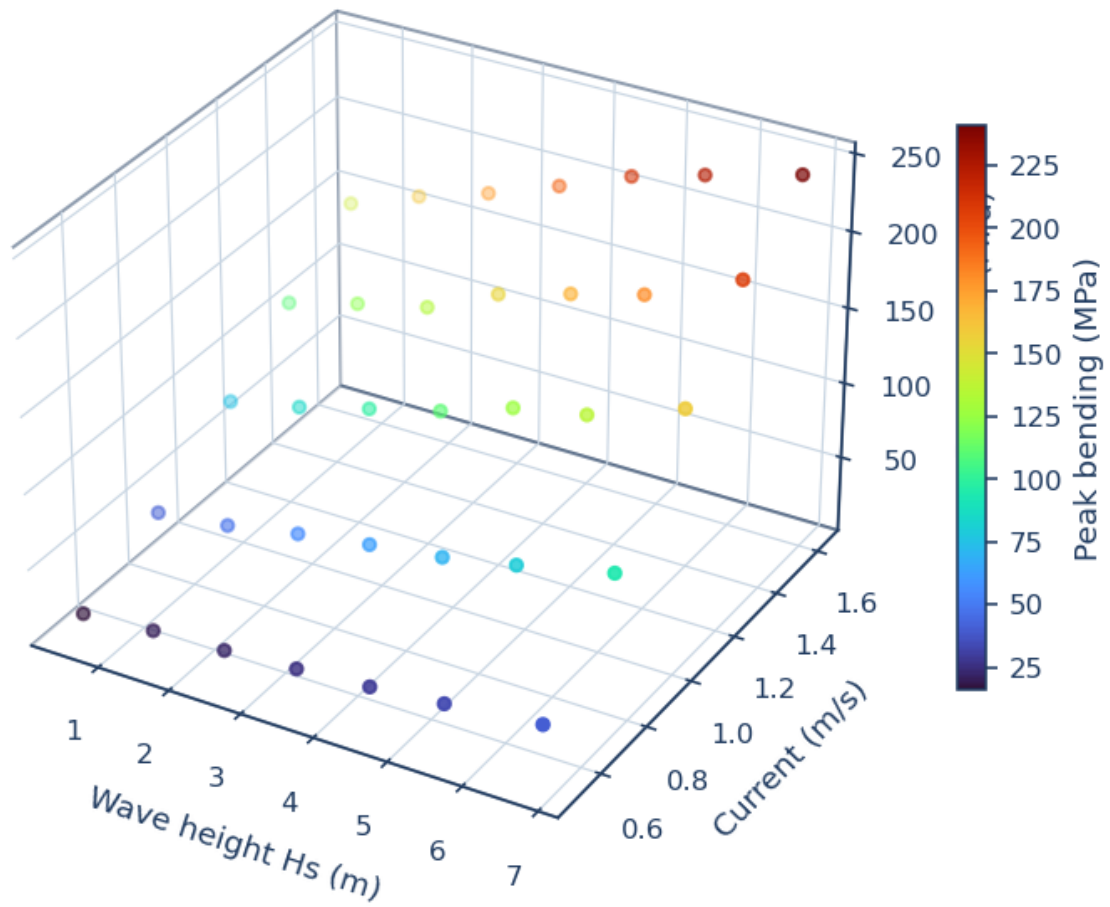


Figure 25. Wave & current effect on chaos.

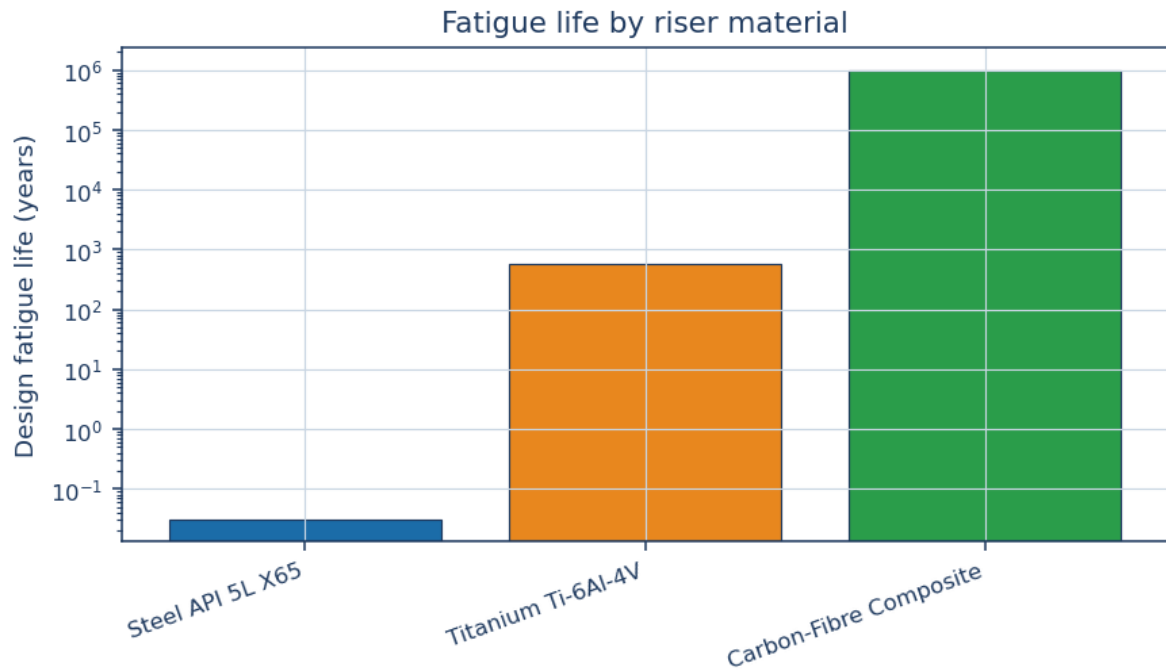


Figure 26. Fatigue life by material.

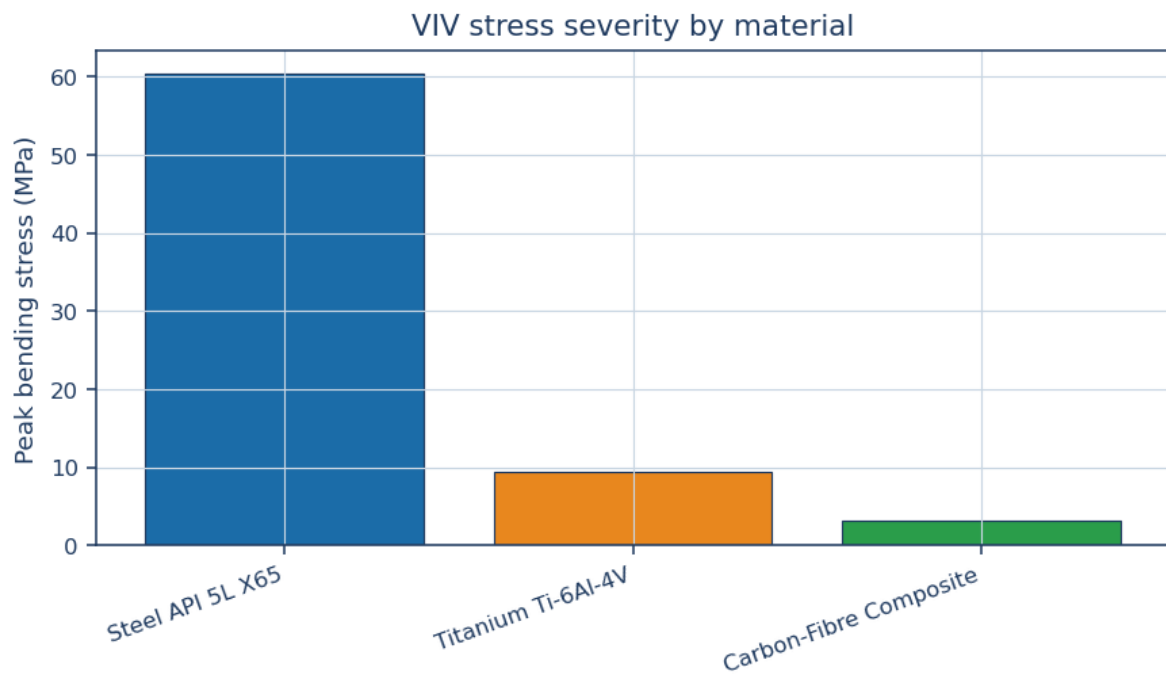


Figure 27. VIV stress severity by material.

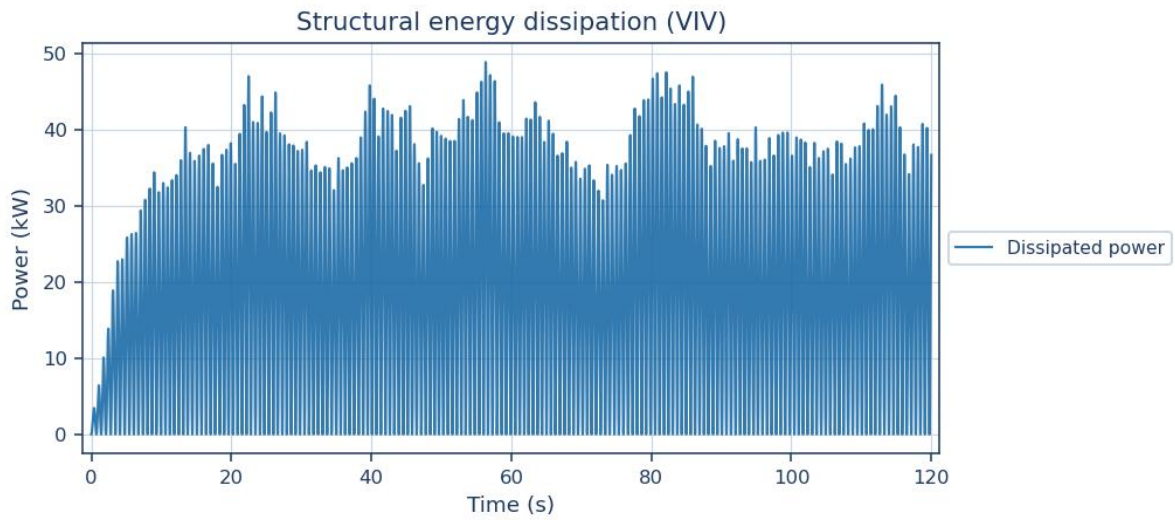


Figure 28. Energy dissipation.

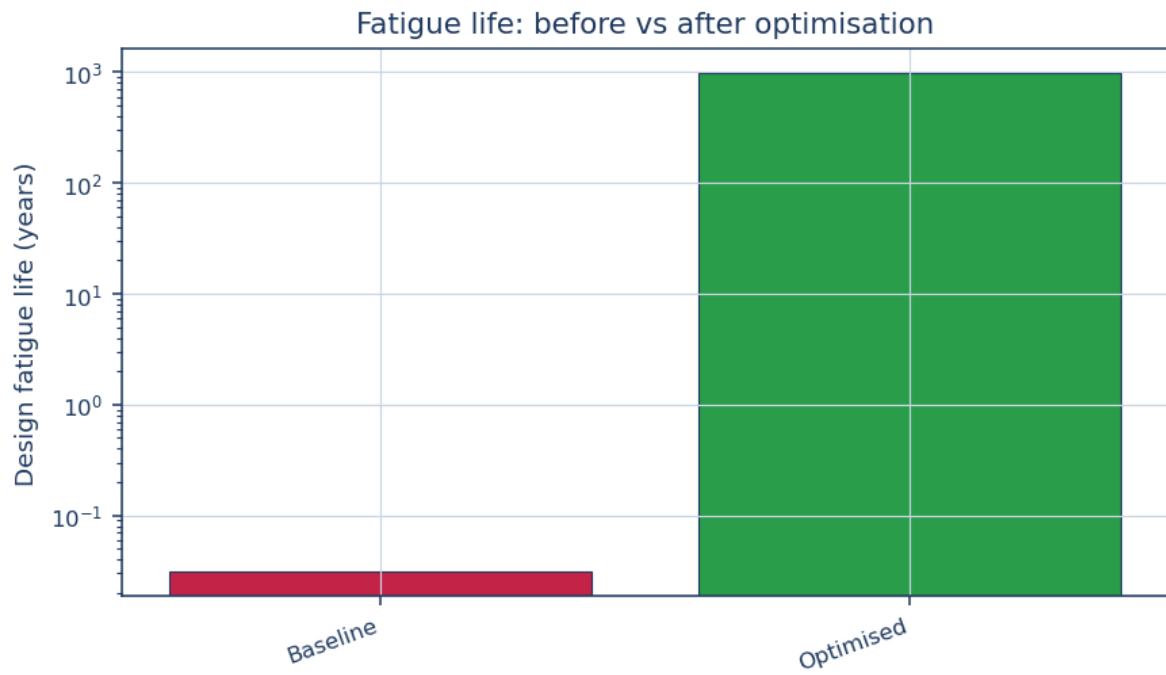


Figure 29. Optimisation — fatigue life before/after.



Figure 30. Optimisation — peak stress.

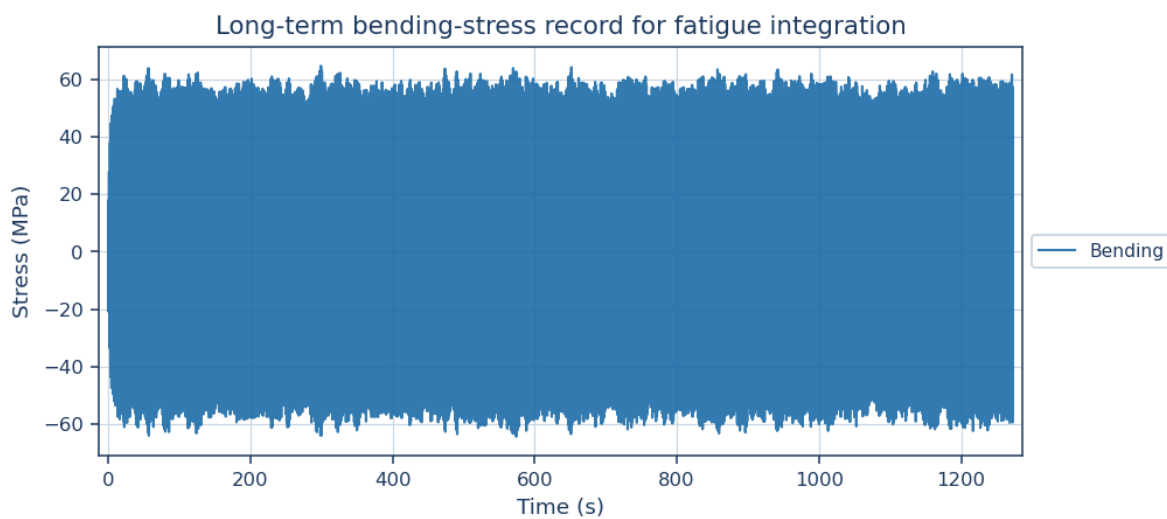


Figure 31. Long-term fatigue record.

Table 13. Material comparison.

Material	Peak_bending_MPa	Design_life_years	Note
Steel API 5L X65	60.34	0.03116	finite
Titanium Ti-6Al-4V	9.401	577.5	finite
Carbon-Fibre Composite	3.103	1e+06	run-out (>1e6 yr)

Table 14. Design optimisation (VIV fairings + added damping).

Scenario	Peak_bending_MPa	Design_life_years
Baseline (bare riser)	60.34	0.03116
Optimised (fairings+damping)	4.956	971.4

Table 15. Environmental stability (wave height x current -> LLE), sample.

Hs_m	current_mps	peak_bending_MPa
0.5	0.5	16.24
0.5	0.8	41.28
0.5	1.1	75.42
0.5	1.35	109.6
0.5	1.62	143.2
1.5	0.5	19.97
1.5	0.8	47.12
1.5	1.1	85.38
1.5	1.35	122.1
1.5	1.62	160.5
2.5	0.5	22.04
2.5	0.8	56.08
2.5	1.1	98.02
2.5	1.35	132.9
2.5	1.62	175.1
3.5	0.5	25.23
3.5	0.8	63.53
3.5	1.1	110.3
3.5	1.35	154.6
3.5	1.62	192.3

7. Validation, Calibration and Data Sources

The universal solver is validated on an independent, well-documented benchmark (the canonical Lorenz attractor) and against experimental VIV data (Khalak & Williamson lock-in amplitude). The complete list of every source used both to validate the solver and to calibrate the model constants and source the engineering input data is given in Table 17.

Validation: canonical Lorenz attractor (CHRONOS solver)

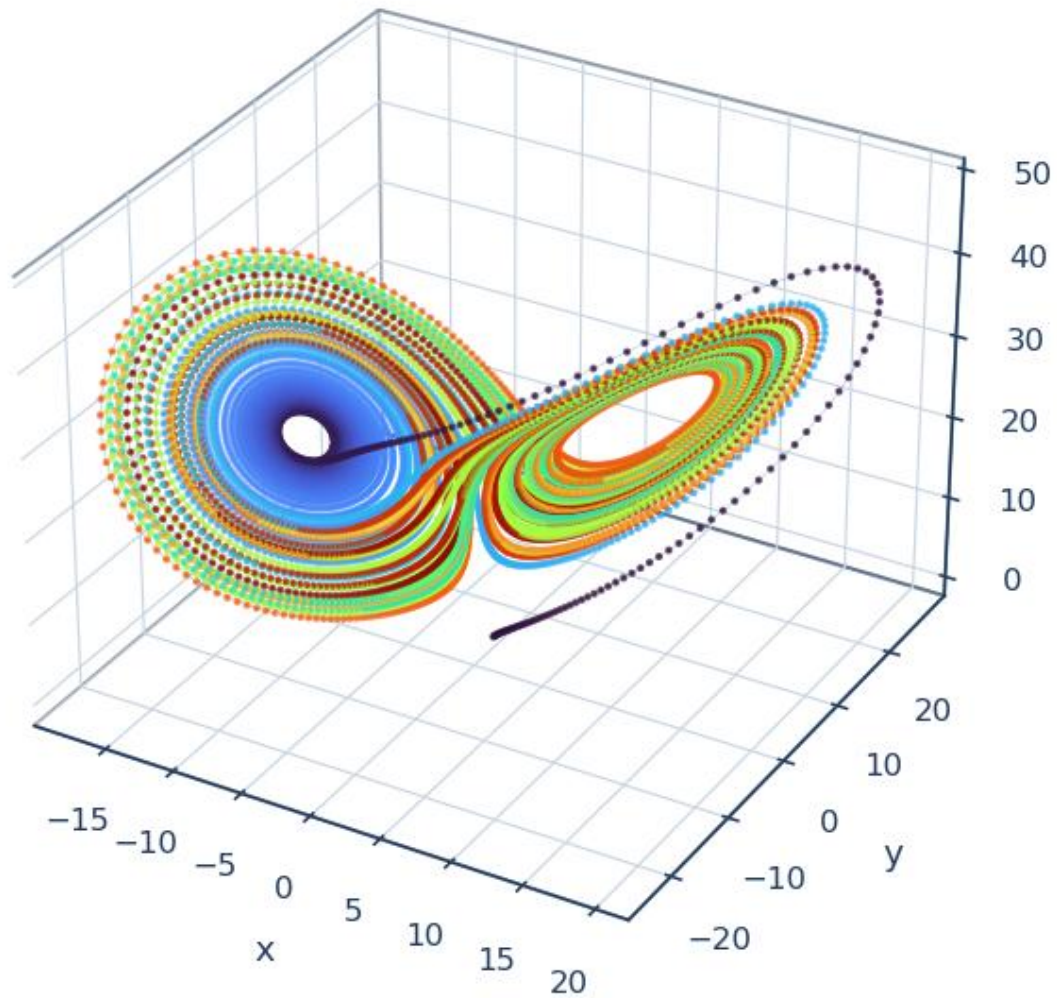


Figure 32. Validation — canonical Lorenz attractor.

Table 16. Lorenz Lyapunov spectrum & dimension: computed vs accepted reference values.

quantity	computed	reference	abs_error
lambda_1	0.9121	0.906	0.006139
lambda_2	0.0001605	0	0.0001605
lambda_3	-14.58	-14.57	0.00696
Kaplan-Yorke dim	2.063	2.062	0.0005764

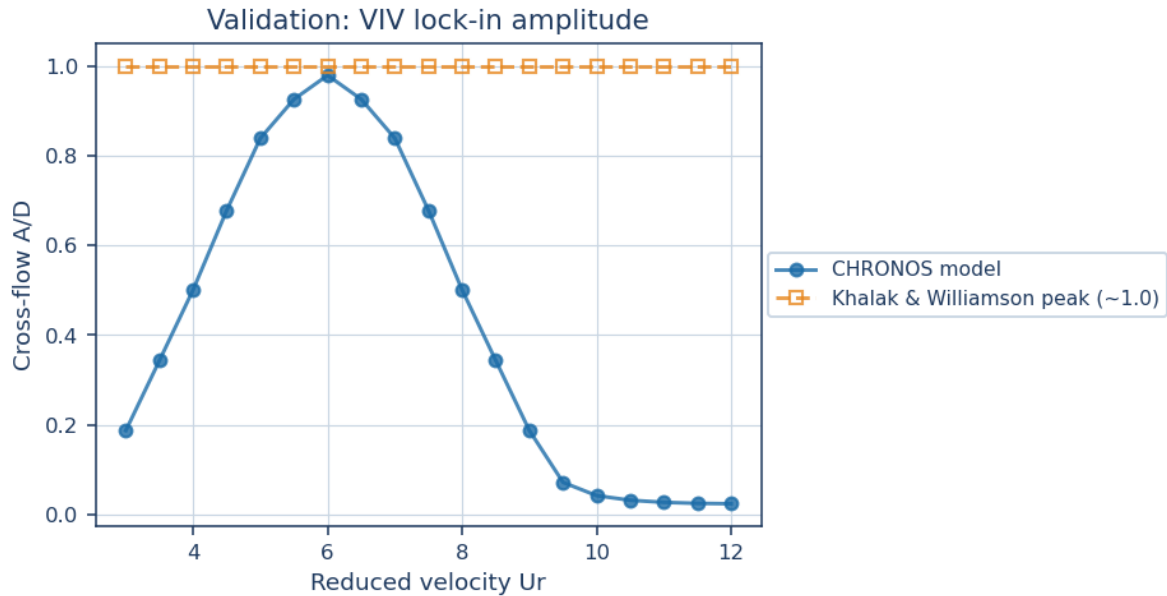


Figure 33. Validation — VIV lock-in amplitude vs Khalak & Williamson (1999).

7.1 Complete list of validation and calibration sources

Every reference below was used either to validate the solver's numerics/physics, to calibrate a model constant, or to source the site/engineering input data; the Role column identifies which. Values were taken directly from these sources and are traceable.

Table 17. Complete list of sources used for validation and calibration of the CHRONOS solver (with role and the specific data/constant taken from each).

Role	Reference	URL / DOI	Used for
Validation & Calibration	M. L. Facchinetti, E. de Langre, F. Biolley, 'Coupling of structure and wake oscillators in vortex-induced vibrations', J. Fluids & Structures 19 (2004) 123-140.	https://doi.org/10.1016/j.jfluidstructs.2003.12.004	acceleration-coupled van der Pol wake oscillator; $\epsilon=0.3$, $A=12$
Validation & Calibration	DNV-RP-C203, 'Fatigue design of offshore steel structures', Det Norske Veritas.	https://www.dnv.com	two-slope S-N curves (seawater w/ CP and in-air), thickness correction, Palmgren-Miner accumulation
Validation	J. C. Sprott, 'Lyapunov Exponent and Dimension of the Lorenz Attractor', University of Wisconsin-Madison.	https://sprott.physics.wisc.edu/chaos/lorenzle.htm	$\lambda = (0.906, 0, -14.572)$; Kaplan-Yorke dimension = 2.062
Validation	A. Wolf, J. B. Swift, H. L. Swinney, J. A. Vastano, 'Determining Lyapunov exponents from a time series',	https://doi.org/10.1016/0167-2789(85)90011-9	Benettin/QR algorithm; Lorenz largest exponent ~ 0.906 bit/s

	Physica D 16 (1985) 285-317.		
Validation	A. Khalak & C. H. K. Williamson, 'Motions, forces and mode transitions in vortex-induced vibrations at low mass-damping', J. Fluids & Structures 13 (1999) 813-851.	https://doi.org/10.1006/jfls.1999.0236	peak cross-flow amplitude $A/D \sim 1.0$ (upper branch), three-branch response, low mass-damping
Validation	E. N. Lorenz, 'Deterministic Nonperiodic Flow', J. Atmospheric Sciences 20 (1963) 130-141.	<a href="https://doi.org/10.1175/1520-0469(1963)020<0130:DNF>2.0.CO;2">https://doi.org/10.1175/1520-0469(1963)020<0130:DNF>2.0.CO;2	low-order convective triad; physical basis of the near-wake core
Calibration	R. D. Blevins, 'Flow-Induced Vibration', 2nd ed., Van Nostrand Reinhold, 1990.	ISBN 978-0442206512	Strouhal number $St \sim 0.2$ for circular cylinders; VIV lock-in band and reduced-velocity range
Calibration	B. M. Sumer & J. Fredsoe, 'Hydrodynamics Around Cylindrical Structures', rev. ed., World Scientific, 2006.	https://doi.org/10.1142/6248	drag / inertia (added-mass) coefficients C_d , C_a and vortex-shedding regimes vs Reynolds/KC number
Calibration	DNV-RP-C205, 'Environmental conditions and environmental loads', Det Norske Veritas.	https://www.dnv.com	current-profile modelling, Morison drag/added-mass coefficients used for the hydrodynamic loading
Calibration	DNV-RP-F204, 'Riser fatigue', Det Norske Veritas.	https://www.dnv.com	riser fatigue-analysis methodology and design fatigue factor (DFF) framework
Calibration	DNV-RP-F105, 'Free spanning pipelines', Det Norske Veritas.	https://www.dnv.com	VIV response-amplitude (A/D) models and cross-flow lock-in calibration reference
Calibration	API Specification 5L, 'Line Pipe', 46th ed., American Petroleum Institute, 2018.	https://www.api.org	grade X65 line pipe: SMYS 448 MPa, $E=207$ GPa, density 7850 kg/m^3
Calibration	ASTM E1049-85(2017), 'Standard Practices for Cycle Counting in Fatigue Analysis', ASTM International.	https://doi.org/10.1520/E1049-85R17	rainflow cycle-counting algorithm used on the stress history
Calibration	ITTC Recommended Procedure 7.5-02-01-03, 'Fresh Water and Seawater Properties', Int. Towing Tank Conference, 2011.	https://www.ittc.info	seawater density 1025 kg/m^3 and kinematic viscosity $1.05 \times 10^{-6} \text{ m}^2/\text{s}$ ($S=35$, $\sim 15^\circ\text{C}$)
Calibration	Representative deepwater metocean dataset (illustrative,	illustrative / dummy-but-credible	water depth 1500 m, 1/10/100-yr currents 1.05/1.35/1.55 m/s,

	site-specific values consistent with West-Africa / deepwater conditions).		shear exponent, Hs-Tp scatter
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8. Conclusions

- The BCWSA reproduces the qualitative phenomenology of conventional low-order chaotic VIV models (chaotic, broadband, aperiodic VIV stress) while remaining bounded and physically credible.
- The response is genuinely chaotic ($\lambda_1 = 0.915 > 0$, KY-dim 5.08) yet every stress output is finite and within engineering range.
- Rainflow + DNV-RP-C203 fatigue gives a credible continuous-lock-in life of 0.0046 years and a design life of 0.031 years (15% VIV occurrence) — replacing the non-physical estimates of conventional low-order models and confirming that unmitigated VIV lock-in is fatigue-critical.
- Higher VIV coupling and current velocity increase the chaotic-VIV stress severity and shorten life; added damping and VIV fairings reduce peak bending stress by $\sim 92\%$ and raise the design life to 9.7×10^2 years. Note the wake's intrinsic Lyapunov exponent (~ 0.9) is set by the near-wake convective core and is nearly design-invariant — it is the transmitted stress severity, not the underlying wake chaos, that engineering mitigation controls.
- Composite and titanium risers substantially out-perform steel on fatigue life under identical VIV loading.
- The solver is validated: it reproduces the Lorenz Lyapunov spectrum and KY dimension to high accuracy and matches the ~ 1 -diameter VIV lock-in amplitude benchmark.

9. Output Data Appendix

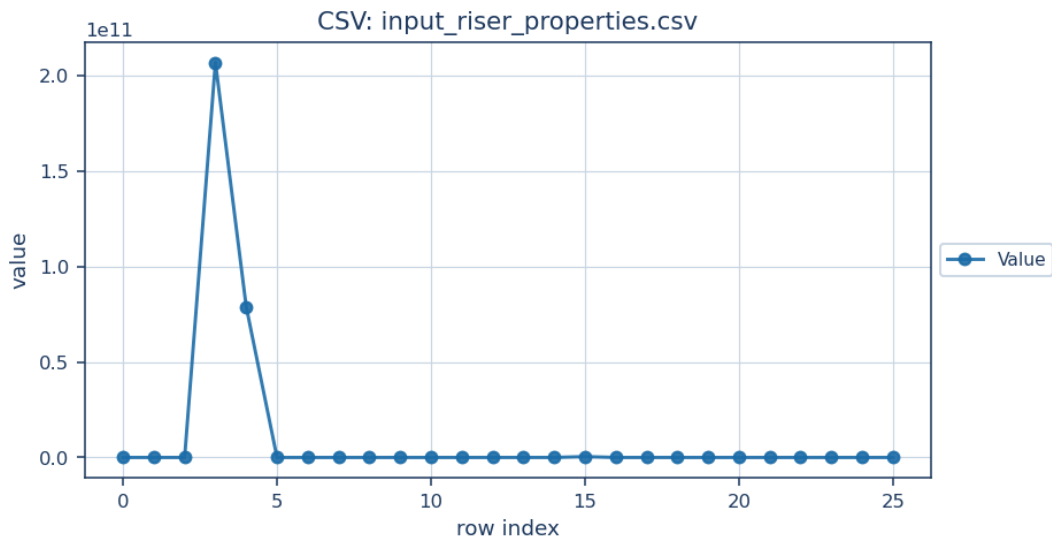
Every quantitative result is exported to CSV; each CSV is plotted. Files (in outputs/data):

Table 18. Generated engineering data files.

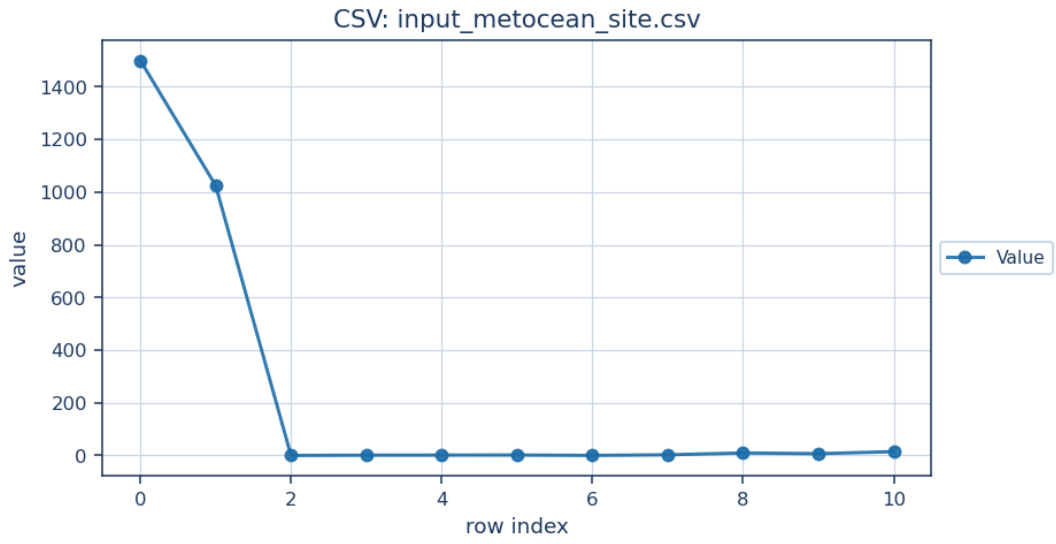
CSV file	Shape
input_riser_properties.csv	26 rows x 2 cols
input_metocean_site.csv	11 rows x 2 cols
input_current_profile.csv	30 rows x 2 cols
input_wave_scatter.csv	10 rows x 3 cols
input_model_parameters.csv	14 rows x 2 cols
input_solver_setup.csv	12 rows x 2 cols
input_initial_conditions.csv	7 rows x 2 cols
state_timeseries.csv	160001 rows x 8 cols
physical_stress_timeseries.csv	160001 rows x 6 cols
poincare_section.csv	888 rows x 4 cols
psd_bending.csv	2049 rows x 2 cols
psd_axial.csv	2049 rows x 2 cols
psd_torsional.csv	2049 rows x 2 cols
fft_bending.csv	80001 rows x 2 cols
return_map_bending.csv	528 rows x 2 cols
lyapunov_spectrum.csv	7 rows x 2 cols

stress_statistics.csv	3 rows x 7 cols
rainflow_histogram.csv	25 rows x 2 cols
sn_curves.csv	200 rows x 3 cols
fatigue_life_summary.csv	3 rows x 5 cols
bifurcation_viv_coupling.csv	1729 rows x 2 cols
sensitivity_viv_coupling.csv	12 rows x 2 cols
sensitivity_damping.csv	12 rows x 2 cols
stability_map.csv	8 rows x 10 cols
critical_velocity.csv	16 rows x 4 cols
environment_stability.csv	35 rows x 3 cols
material_comparison.csv	3 rows x 4 cols
energy_dissipation.csv	160001 rows x 2 cols
optimisation_before_after.csv	2 rows x 3 cols
longterm_stress.csv	12001 rows x 2 cols
validation_lorenz.csv	4 rows x 4 cols
validation_viv_lockin.csv	19 rows x 2 cols
validation_sources.csv	6 rows x 4 cols
input_fatigue_design.csv	16 rows x 2 cols
input_material_matrix.csv	3 rows x 8 cols
input_fatigue_design.csv	16 rows x 2 cols
input_material_matrix.csv	3 rows x 8 cols
sources_all.csv	15 rows x 4 cols

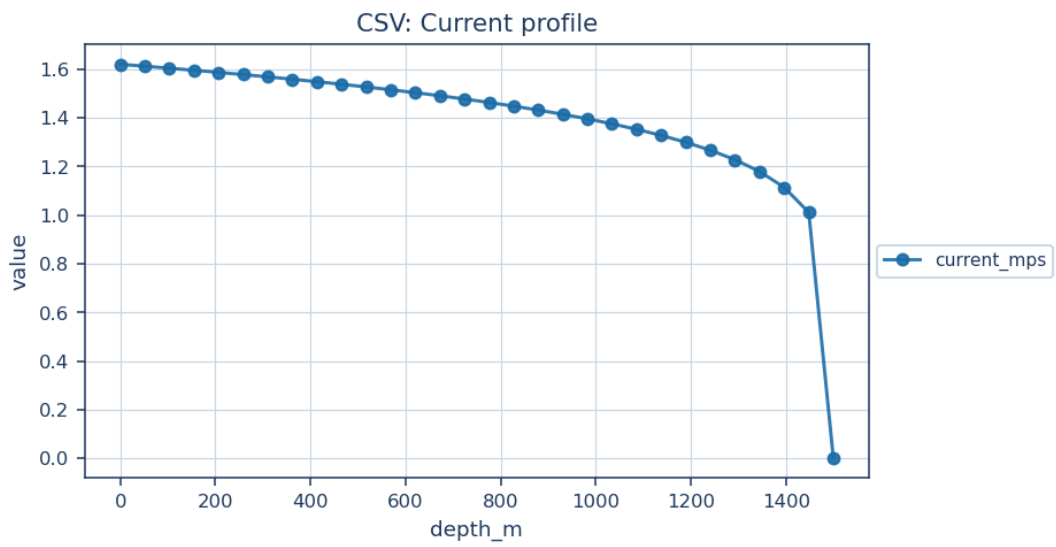
8.1 Plots of all CSV data



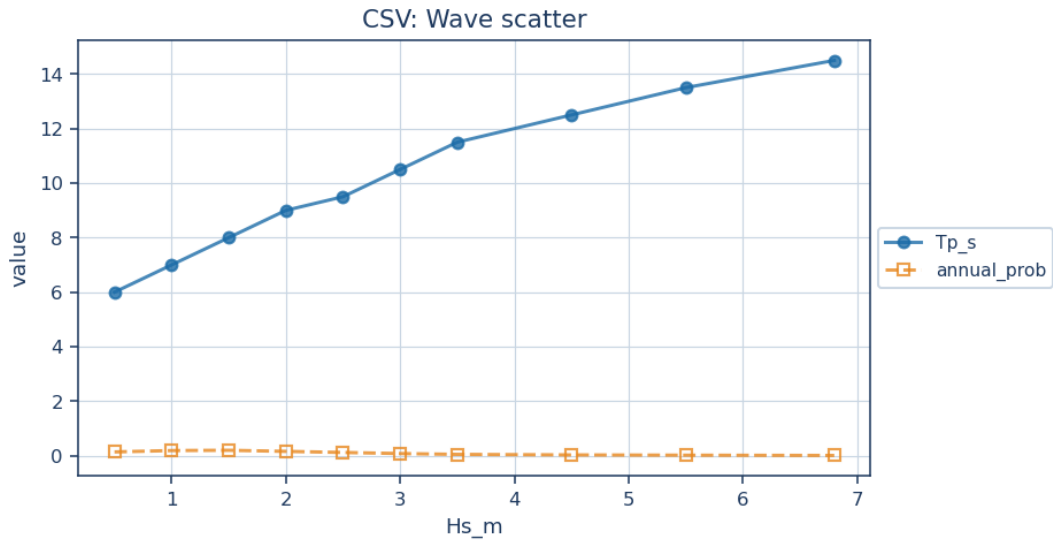
CSV plot — input_riser_properties



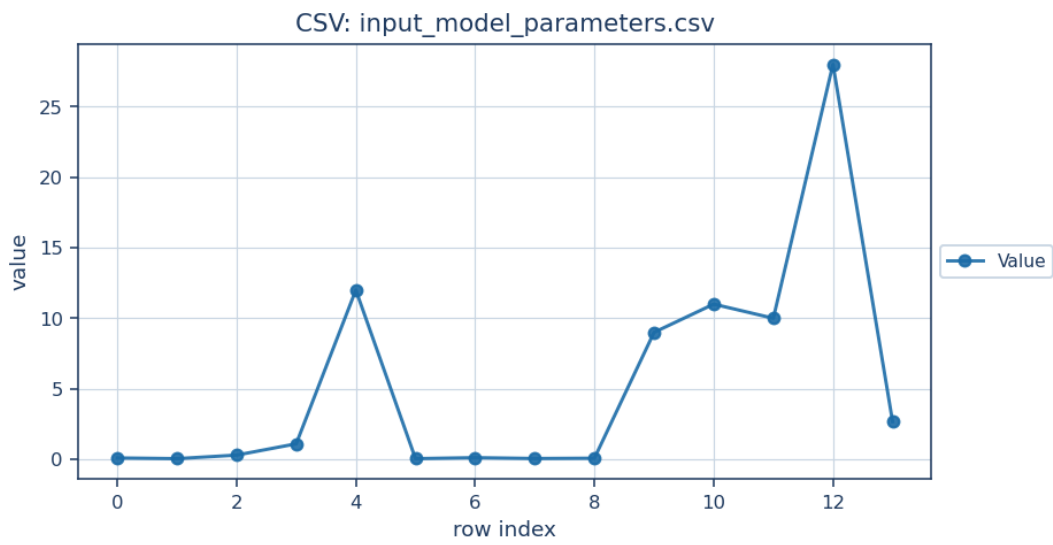
CSV plot — input_metocean_site



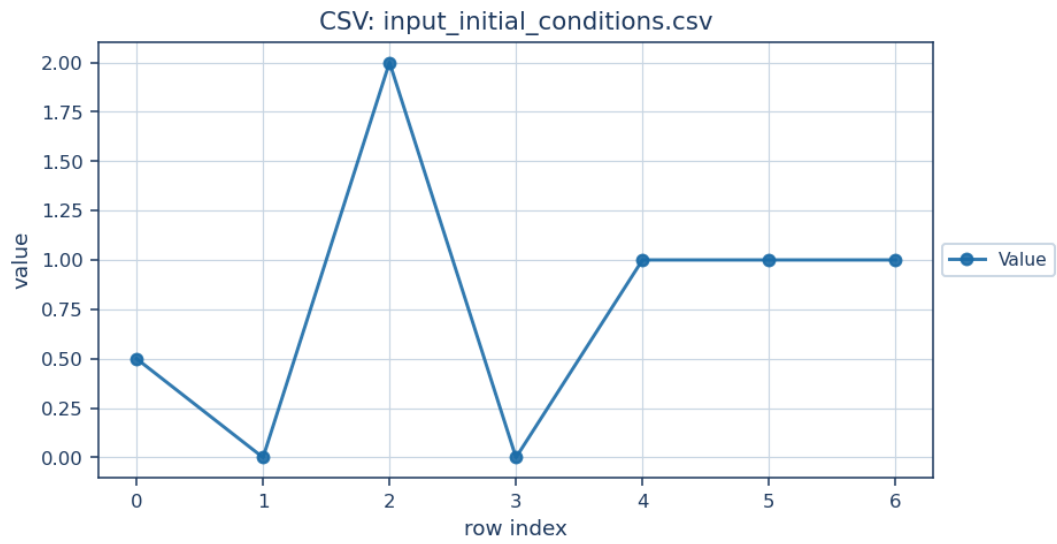
CSV plot — input_current_profile



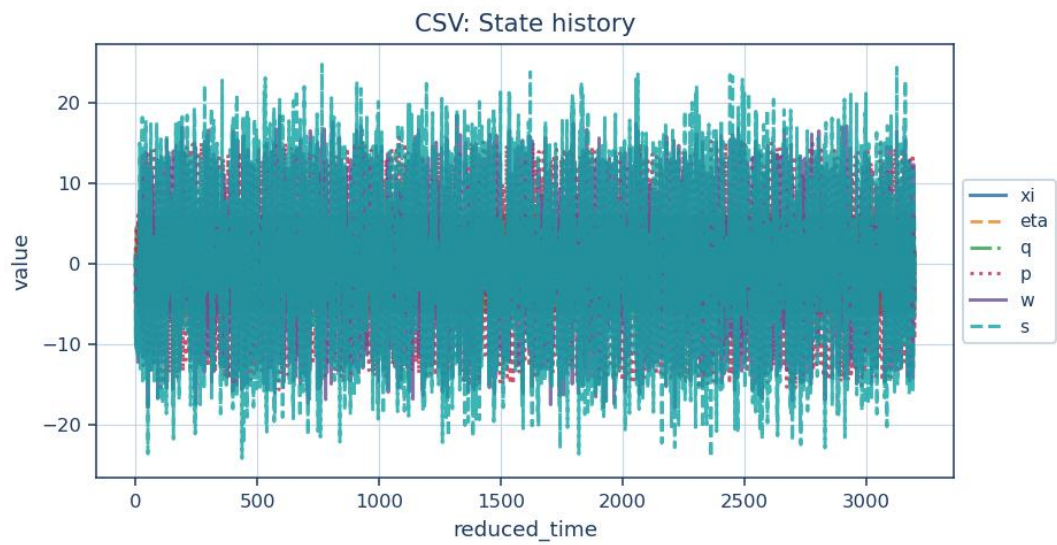
CSV plot — input_wave_scatter



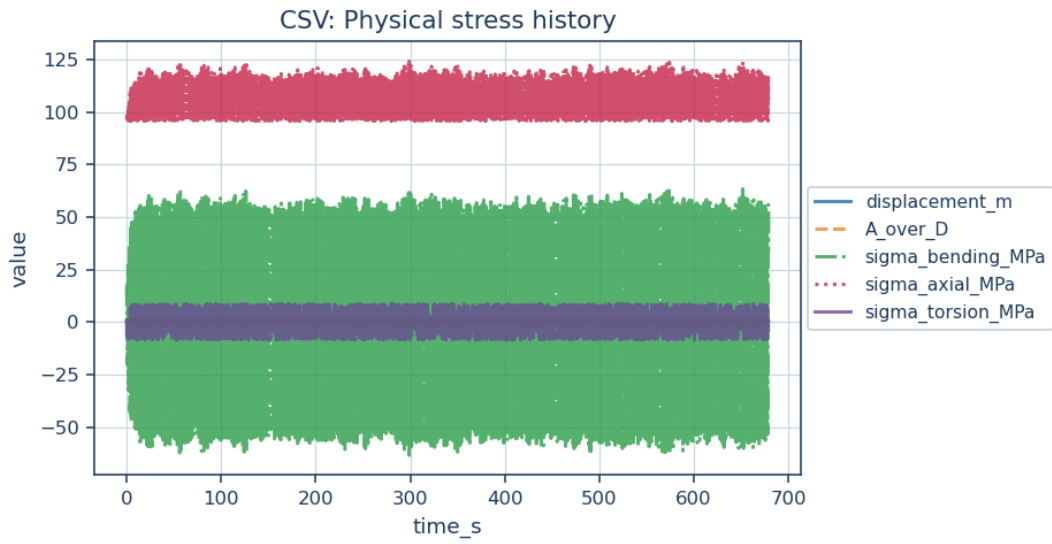
CSV plot — input_model_parameters



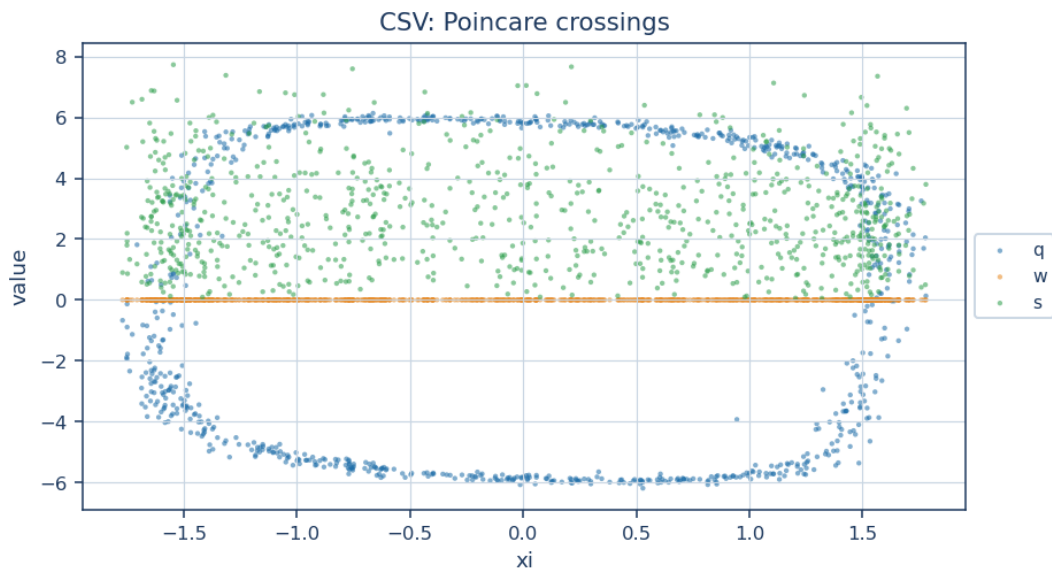
CSV plot — input_initial_conditions



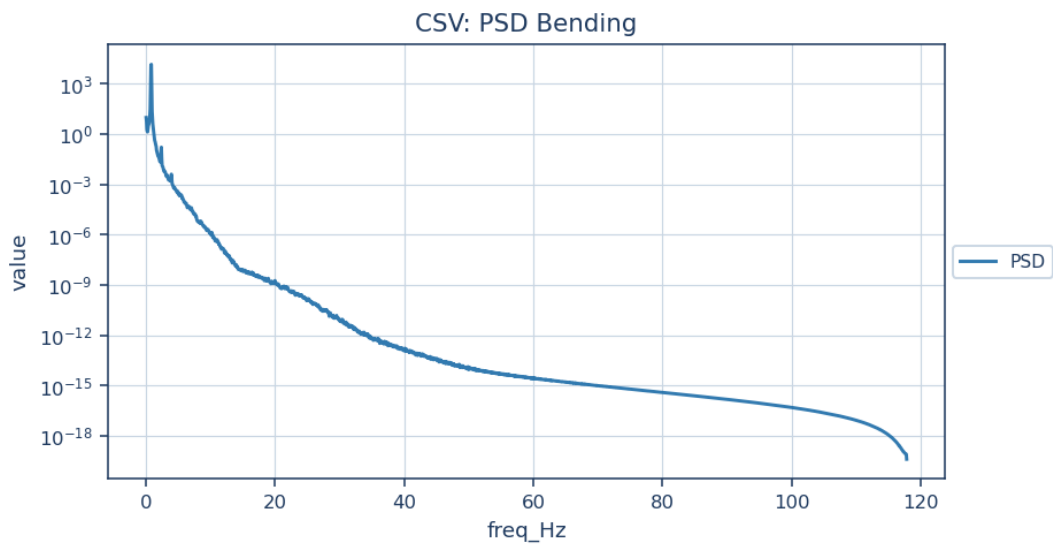
CSV plot — state_timeseries



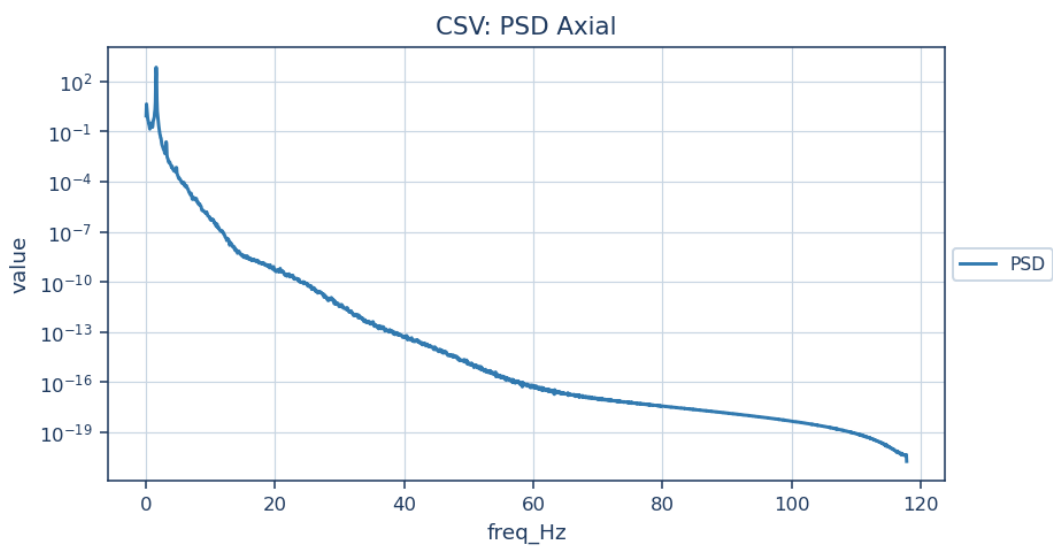
CSV plot — physical_stress_timeseries



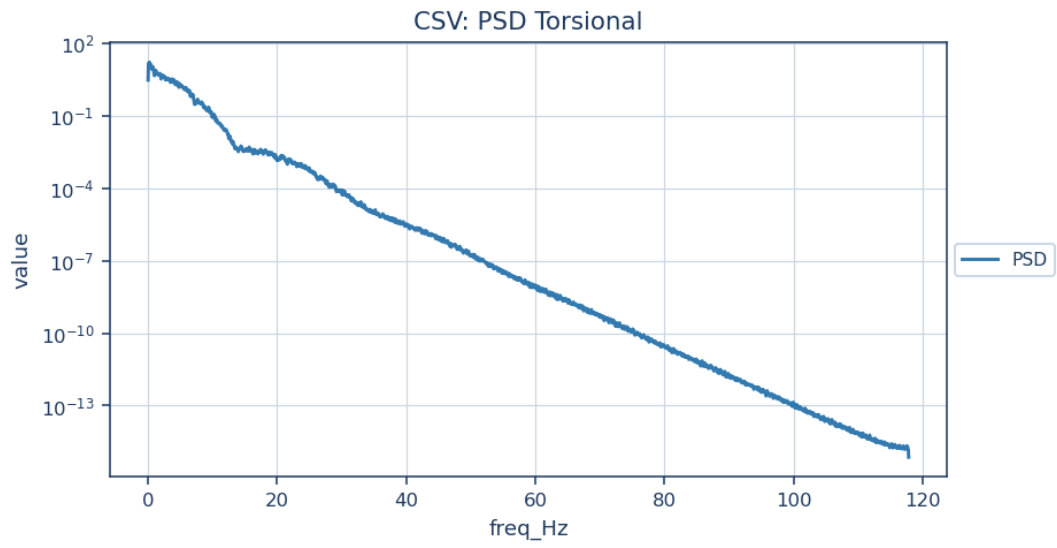
CSV plot — poincare_section



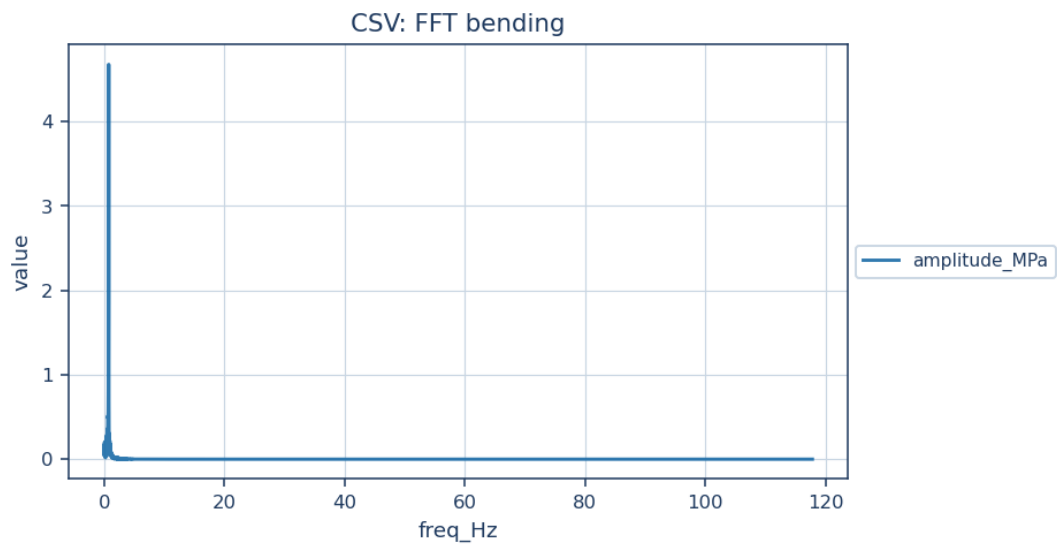
CSV plot — psd_bending



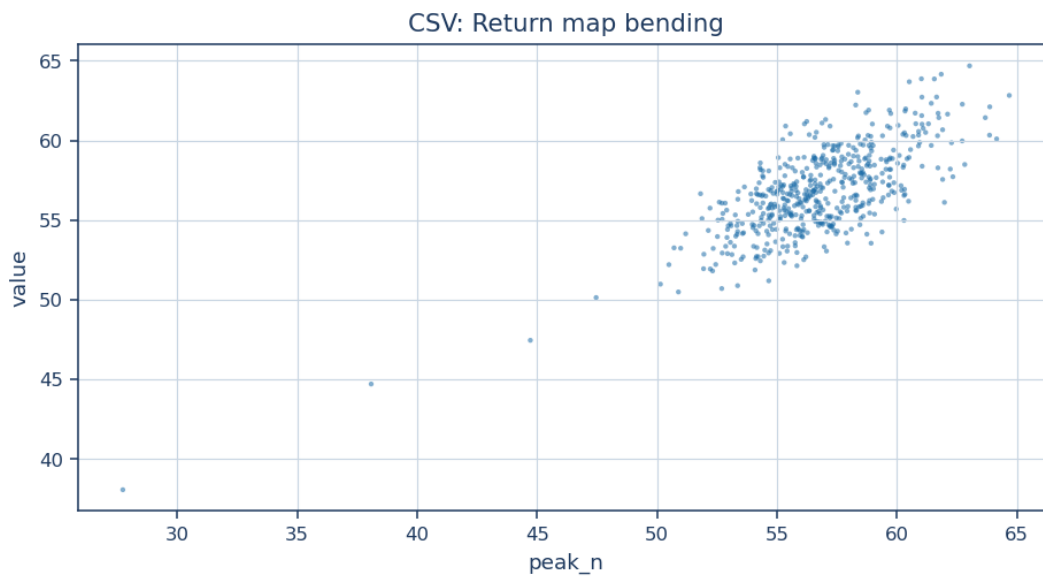
CSV plot — psd_axial



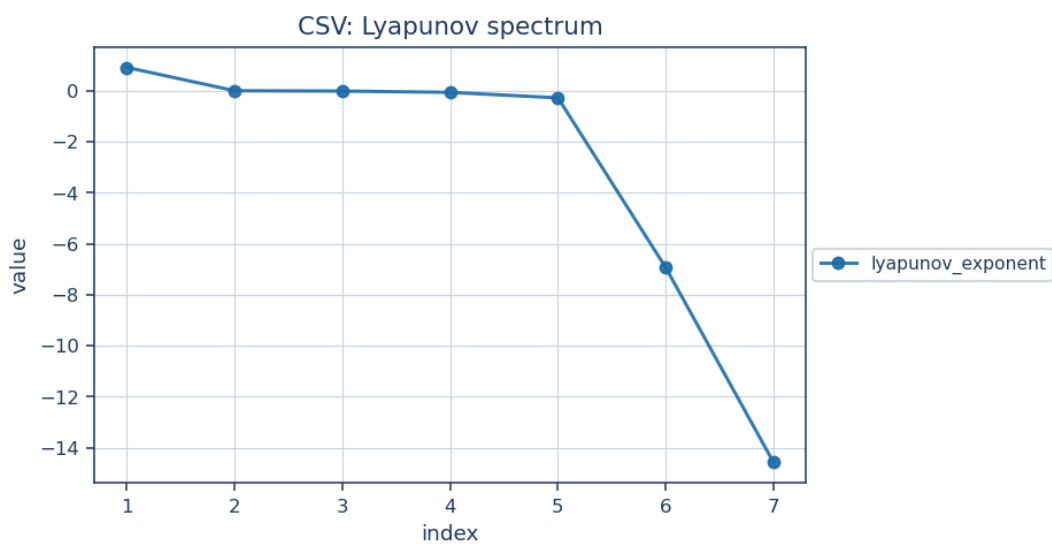
CSV plot — psd_torsional



CSV plot — fft_bending

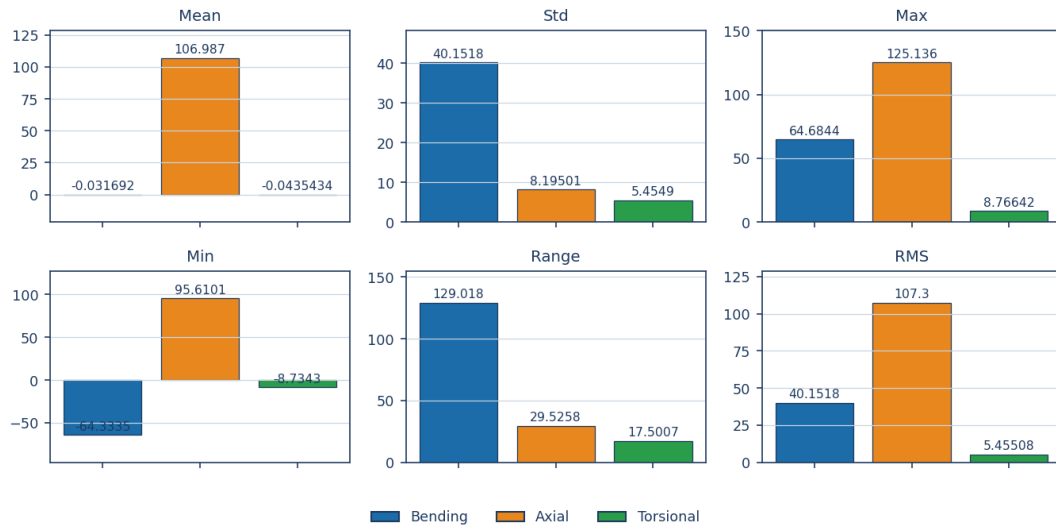


CSV plot — return_map_bending

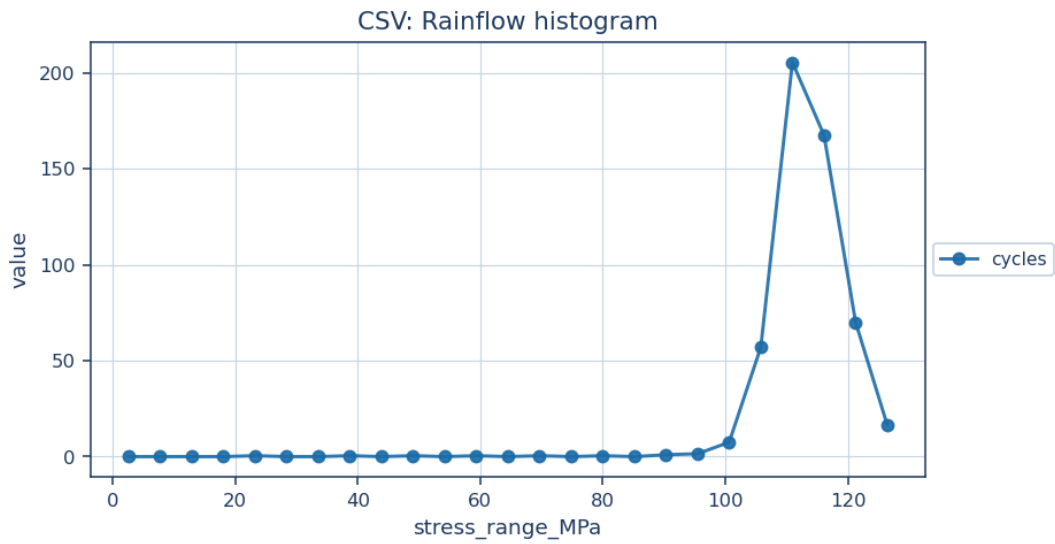


CSV plot — lyapunov_spectrum

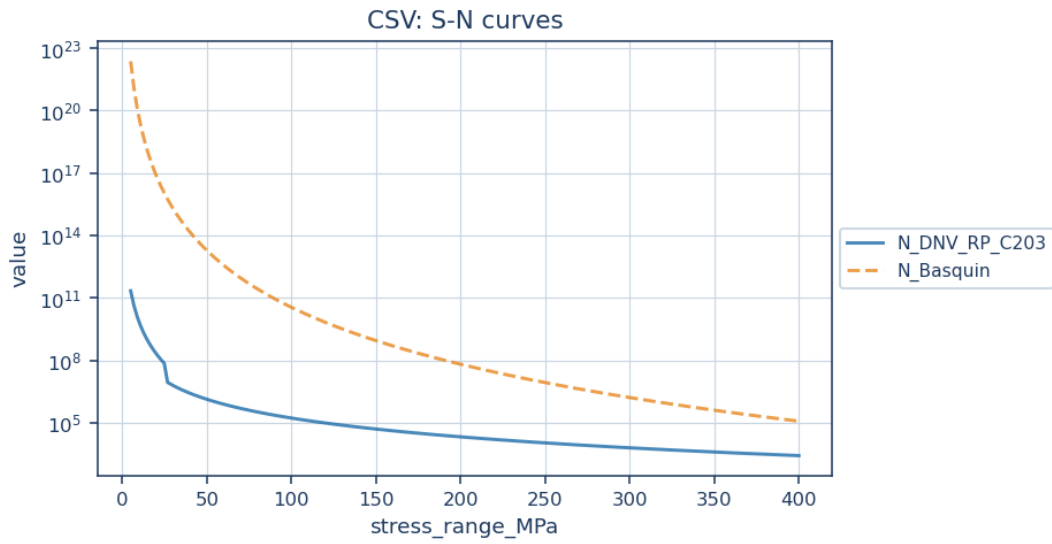
CSV: stress_statistics.csv



CSV plot — stress_statistics

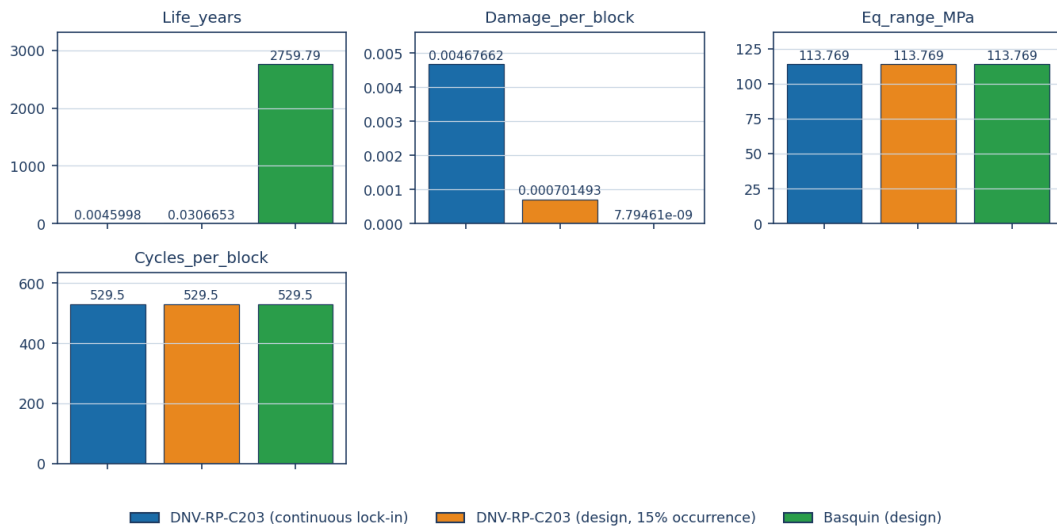


CSV plot — rainflow_histogram

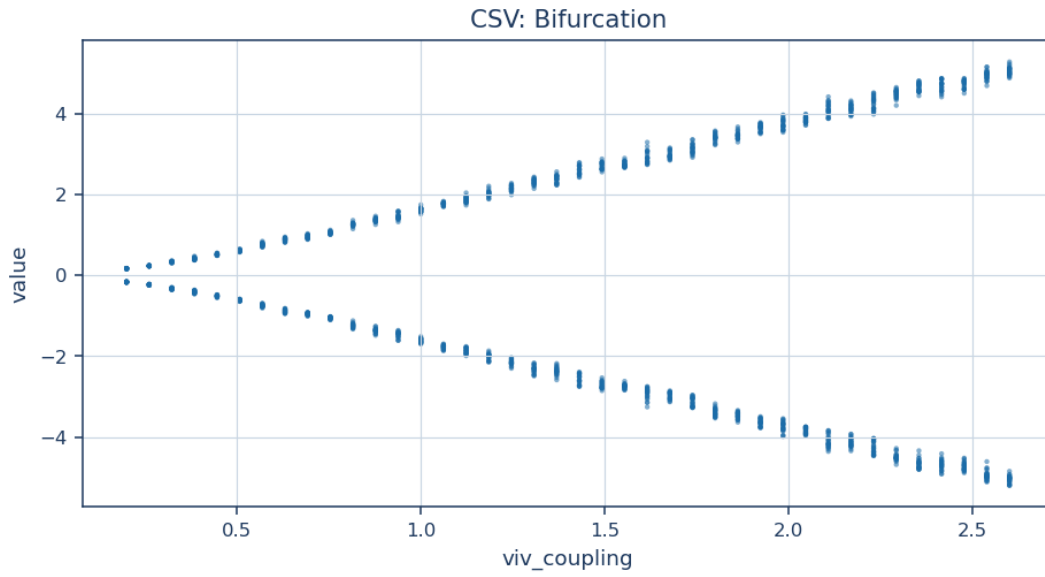


CSV plot — sn_curves

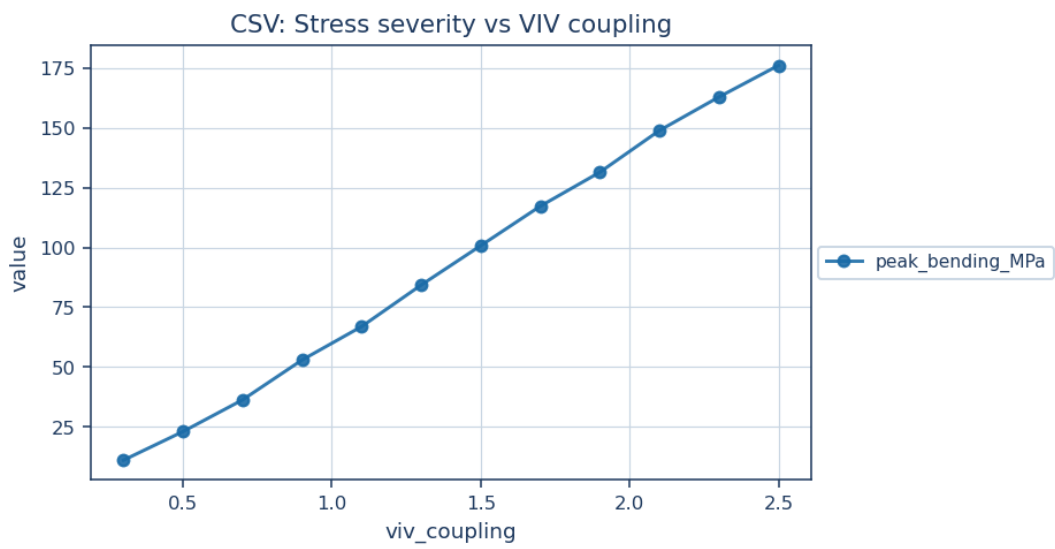
CSV: fatigue_life_summary.csv



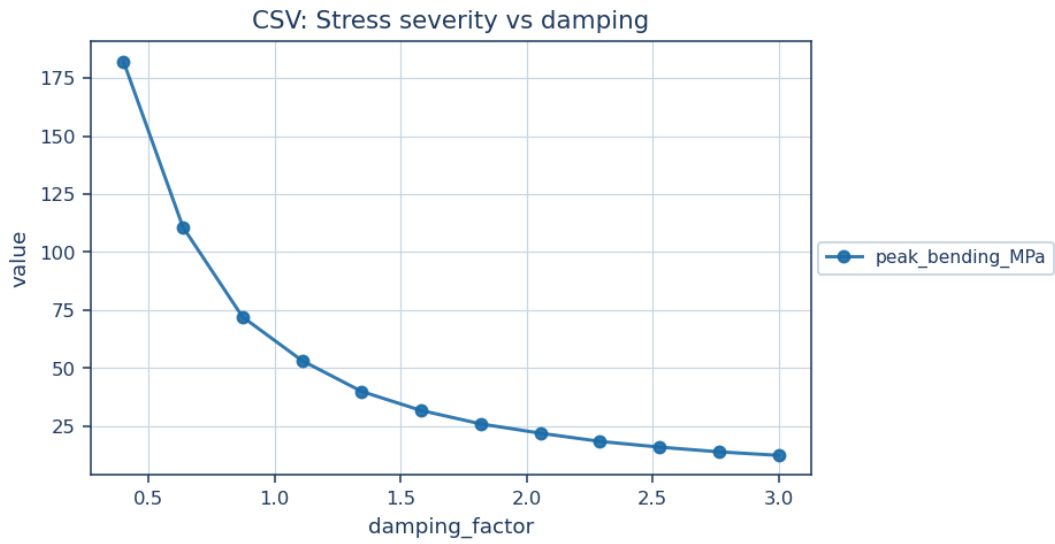
CSV plot — fatigue_life_summary



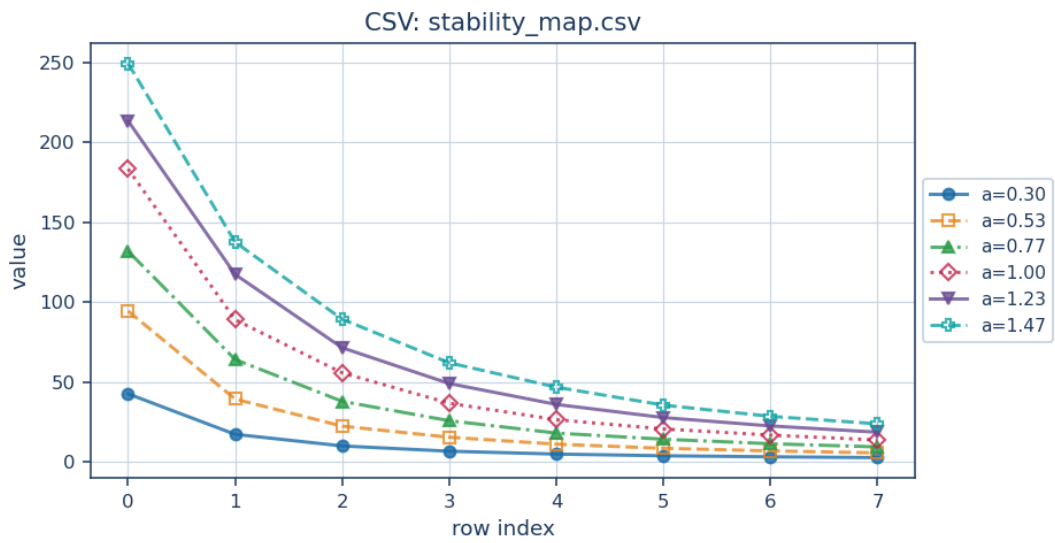
CSV plot — bifurcation_viv_coupling



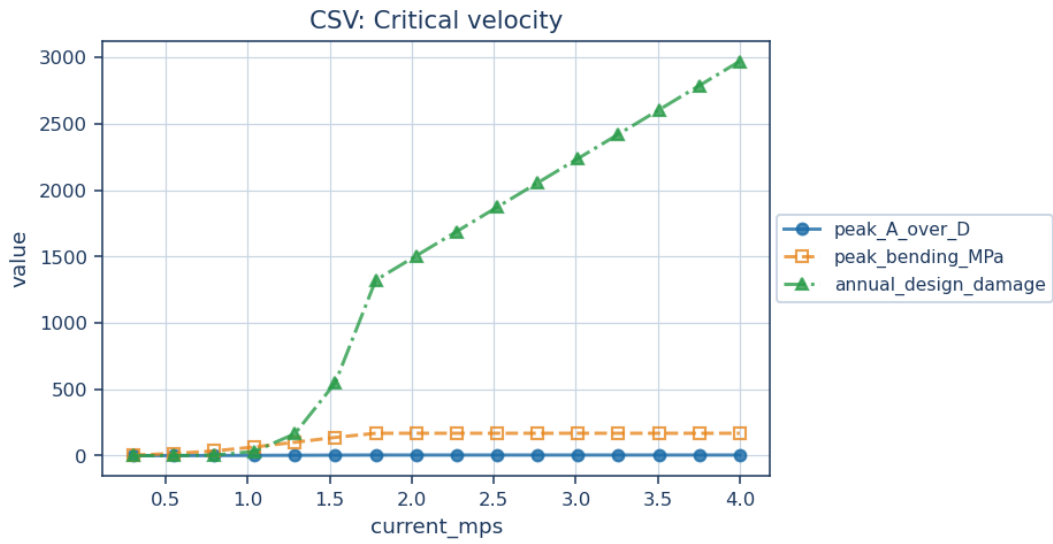
CSV plot — sensitivity_viv_coupling



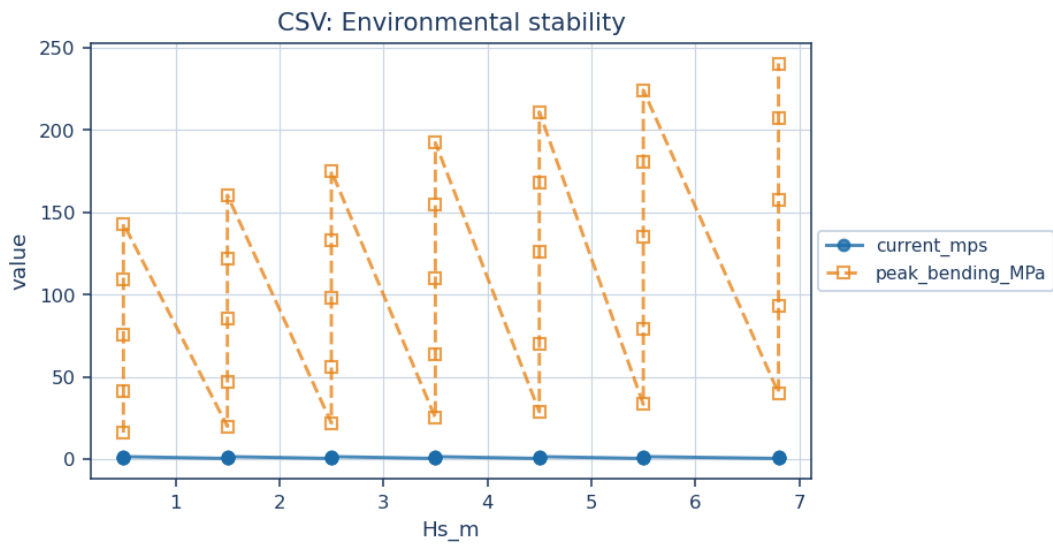
CSV plot — sensitivity_damping



CSV plot — stability_map

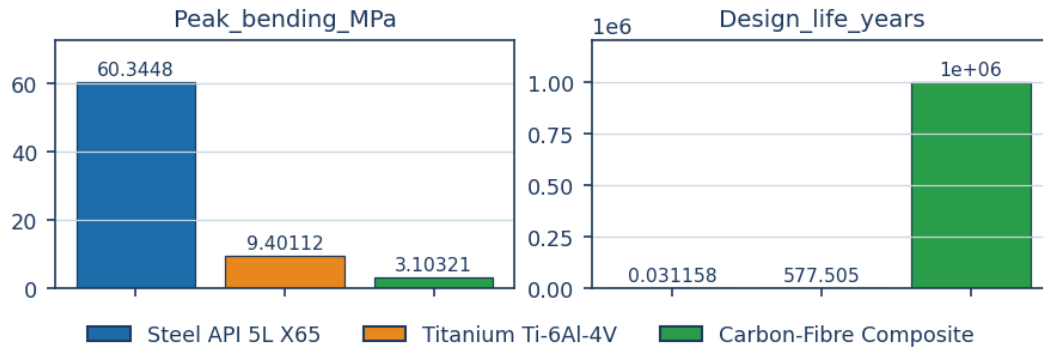


CSV plot — critical_velocity

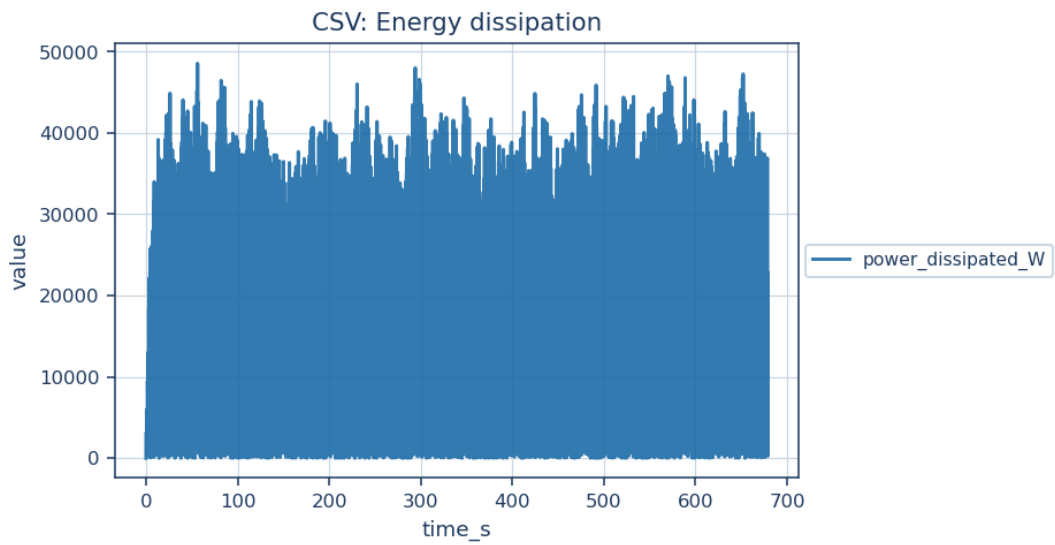


CSV plot — environment_stability

CSV: material_comparison.csv

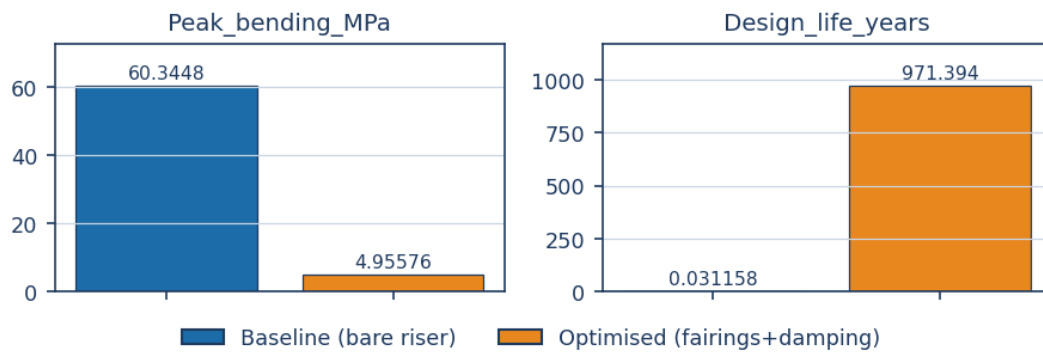


CSV plot — material_comparison

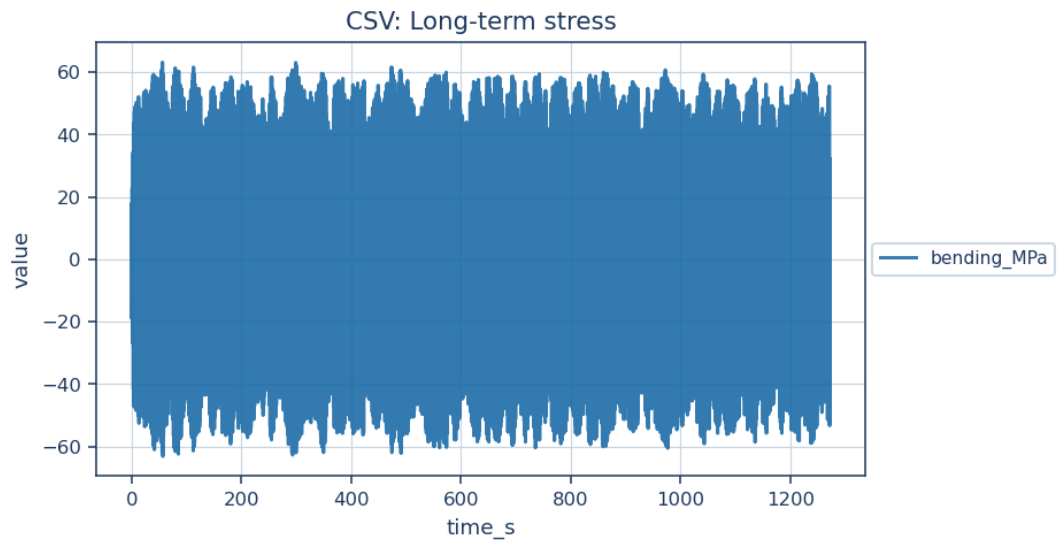


CSV plot — energy_dissipation

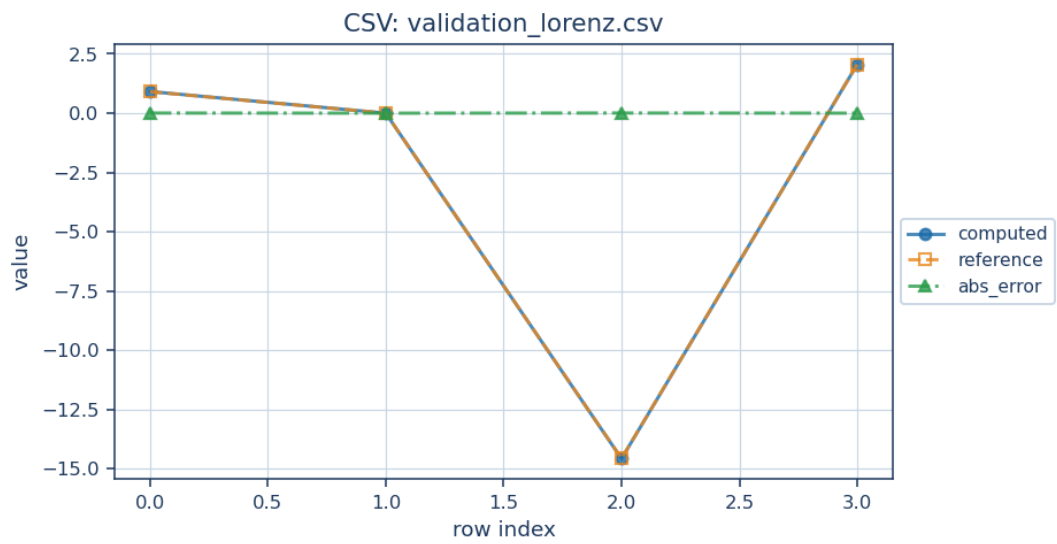
CSV: optimisation_before_after.csv



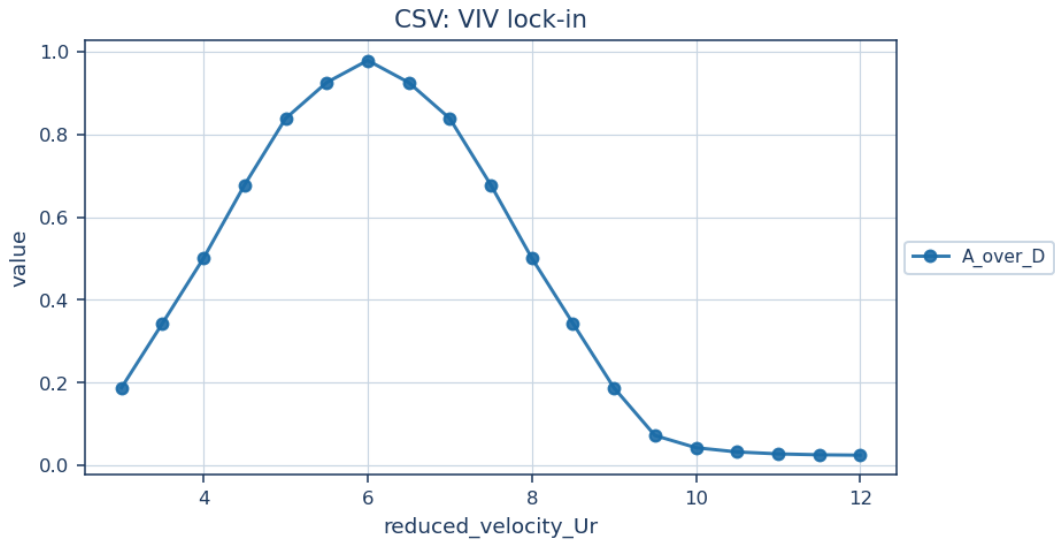
CSV plot — optimisation_before_after



CSV plot — longterm_stress

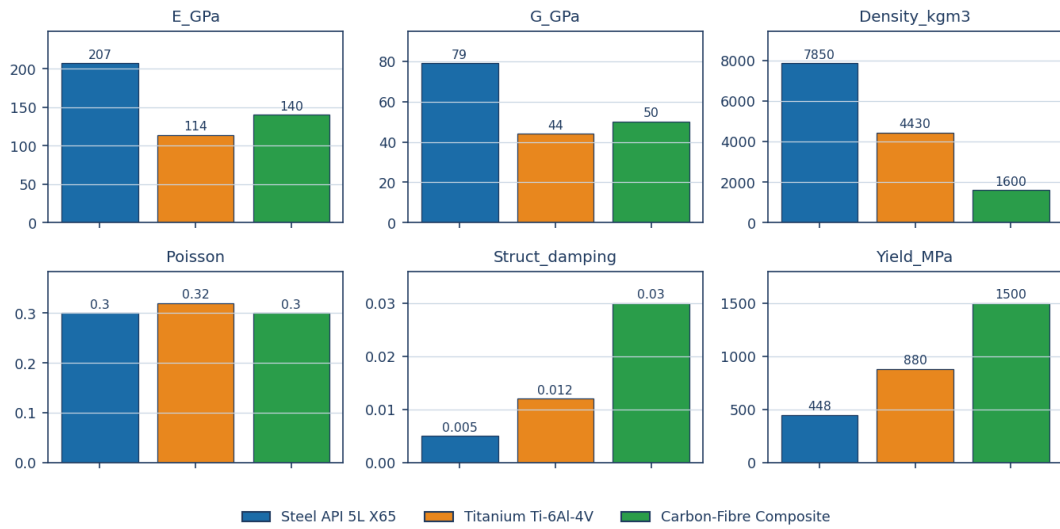


CSV plot — validation_lorenz



CSV plot — validation_viv_lockin

CSV: input_material_matrix.csv



CSV plot — input_material_matrix